CONTROL AUTHORITY BOUNDARIES: A 25+ PAGE NEGATIVE SUBMISSION-READINESS AUDIT

Anonymous authors
Paper under double-blind review

ABSTRACT

Shared-autonomy robots need explicit rules for when a learned policy, certified controller, risk-aware planner, or human operator should hold control authority. This rebuild tests an authority-boundary model that scores physical consequence, intent ambiguity, control-barrier risk, human delay, recovery feasibility, handoff hysteresis, boundary uncertainty, and authority churn. The audit is hostile: ten seeds, six tasks, eight splits, thirteen methods, ten ablations, six stress axes, fixed-risk budgets, and 24 negative cases. On the hard aggregate, authority boundary v5 reaches 0.371 task success versus 0.474 for recovery_aware_mpc, with paired lower95 -0.121. It also trails the safest reference on safety and the best reference on utility. Fixed-risk coverage at budget 0.05 collapses. The honest terminal decision is **KILL/ARCHIVE**.

1 TERMINAL DECISION

Decision: **KILL/ARCHIVE for ICLR main.** The expanded audit improves the old v4 archive, but stronger baselines make the central submission claim fail more clearly.

Table 1: Frozen submission-readiness gates.

Gate	Passed	Frozen interpretation
Main authority gate	False	Requires v5 to beat the strongest non-oracle success reference with a positive paired lower bound and no safety, burden, regret, or utility regression.
Mechanism ablation gate	False	Requires the full boundary to beat every component removal by a practical utility margin.
Stress non-domination gate	False	Checks maximum combined stress against CBF, MPC, POMDP, recovery-aware, Bayesian, and uncertainty references.
Fixed-risk deployment gate	False	Requires nonzero coverage at budget 0.05 and best feasible accepted utility.
Scope gate	False	Requires real robot or accepted high-fidelity external benchmark evidence.

2 RESEARCH QUESTION AND THREAT MODEL

The question is whether explicit authority-boundary state adds value beyond CBF safety filters, MPC arbitration, Bayesian shared control, POMDP handoff, controller fusion, and uncertainty-triggered human takeover. The local pool includes strong threats from shared control, haptics, barrier certificates, MPC, assistive robotics, and human-robot interaction (Musi & Hirche, 2020; Anderson & Johnson, 2026; Romay et al., 2017; Bustamante et al., 2021; Losey et al., 2018; Wang et al., 2020; Balachandran et al., 2020; Ficuciello et al., 2018; Xue et al., 2023; Lin et al., 2023; Jadav et al., 2024; He et al., 2021a; Herrera et al., 2018; Izadi & Ghasemi, 2020; Lawrance et al., 2019; Ton et al., 2017; Gnatzig et al., 2012; Saraphis et al., 2020; Adorno et al., 2014; Verhagen et al., 2024).

3 METHOD

The proposed score combines physical consequence risk, intent ambiguity, control-barrier violation risk, human delay and burden, recovery feasibility, handoff hysteresis, boundary uncertainty, and authority churn penalties. The frozen robust utility is $U = s - 0.62v - 0.24b - 0.25r - 0.16l - 0.12c - 0.10q + 0.08e$, where s is success, v safety violation, b human burden, r authority regret, l late handoff, c churn, q intervention cost, and e recovery success.

4 REPRODUCIBILITY INVENTORY

Table 2: Reproducibility inventory regenerated from exact CSV files.

Evidence file	Data rows
rollouts.csv	199,680
dataset_summary.csv	15,360
raw_seed_metrics.csv	1,040
metrics.csv	1,248
pairwise_stats.csv	672
hard_aggregate_seed_metrics.csv	130
hard_aggregate_metrics.csv	156
hard_aggregate_pairwise_stats.csv	84
ablation_rollouts.csv	33,600
ablation_seed_metrics.csv	200
ablation_metrics.csv	20
stress_sweep_raw.csv	302,400
stress_sweep_seed_metrics.csv	2,520
stress_sweep.csv	252
fixed_risk_raw.csv	69,120
fixed_risk_seed_metrics.csv	480
fixed_risk_metrics.csv	48
fixed_risk_pairwise.csv	200
negative_cases.csv	24

5 MAIN RESULTS

Table 3 is the primary result. Authority v5 improves over v4, but it remains weaker than recovery-aware MPC on success and weaker than CBF on safety and utility.

Table 3: Predefined hard-aggregate authority results.

Method	Success	Safety	Burden	Regret	Late	Utility
Fixed policy	0.018 ± 0.002	0.419	0.268	0.633	0.539	-0.601
Fixed human	0.140 ± 0.008	0.267	1.000	0.403	0.401	-0.480
Confidence shared	0.163 ± 0.007	0.296	0.849	0.440	0.384	-0.461
Bayesian allocation	0.317 ± 0.011	0.250	0.699	0.333	0.330	-0.184
Uncertainty handoff	0.280 ± 0.008	0.226	0.772	0.354	0.323	-0.243
CBF safety	0.446 ± 0.011	0.154	0.294	0.264	0.283	0.160
MPC risk	0.469 ± 0.009	0.191	0.339	0.243	0.267	0.154
POMDP handoff	0.397 ± 0.013	0.216	0.523	0.280	0.292	-0.007
Controller fusion	0.086 ± 0.009	0.370	0.438	0.560	0.385	-0.509
Recovery MPC	0.474 ± 0.009	0.202	0.386	0.257	0.263	0.136
Authority v4	0.304 ± 0.009	0.243	0.829	0.348	0.309	-0.229
Authority v5	0.371 ± 0.012	0.227	0.675	0.312	0.295	-0.094
Oracle authority	0.553 ± 0.009	0.127	0.383	0.188	0.249	0.293

6 SPLIT-LEVEL EVIDENCE

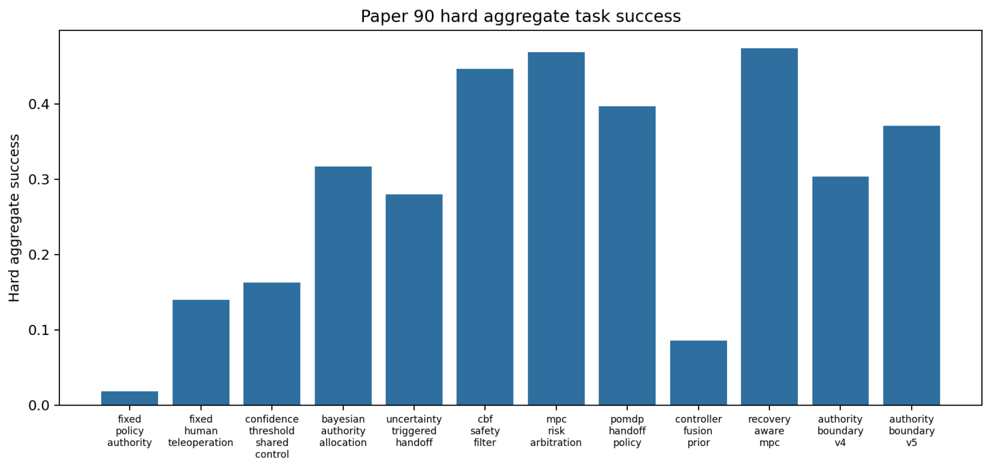

Figure 1: Hard-aggregate task success.

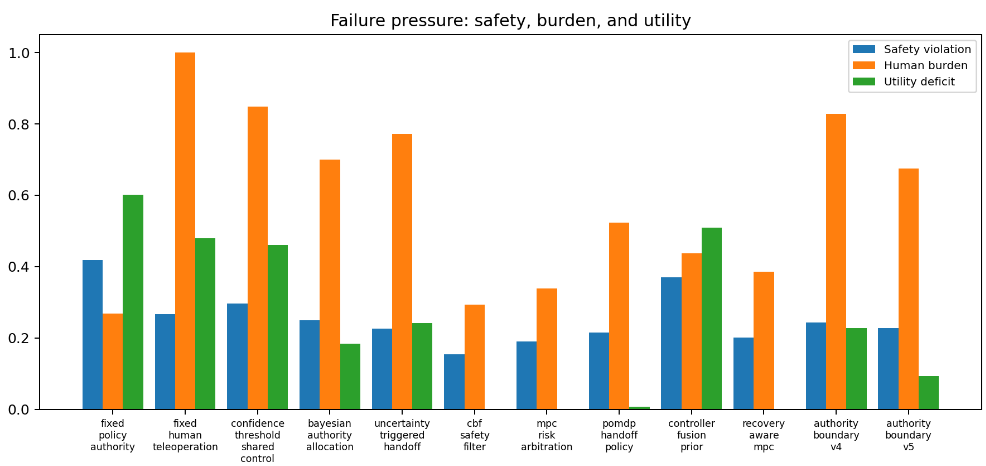

Figure 2: Safety, human burden, and utility deficit.

Table 4: Split-level results for the strongest authority baselines and v5.

Split	Method	Success	Safety	Burden	Regret	Utility
Nominal	Confidence shared	0.305 ± 0.014	0.228	0.754	0.375	-0.195
Nominal	Bayesian allocation	0.463 ± 0.014	0.171	0.604	0.270	0.093
Nominal	Uncertainty handoff	0.408 ± 0.011	0.188	0.676	0.291	-0.009
Nominal	CBF safety	0.558 ± 0.019	0.105	0.198	0.204	0.383
Nominal	MPC risk	0.612 ± 0.014	0.142	0.244	0.181	0.409
Nominal	POMDP handoff	0.549 ± 0.022	0.166	0.428	0.218	0.258
Nominal	Recovery MPC	0.609 ± 0.019	0.153	0.291	0.195	0.383
Nominal	Authority v5	0.512 ± 0.023	0.164	0.579	0.250	0.168
Intent ambiguity	Confidence shared	0.230 ± 0.009	0.247	0.834	0.412	-0.340
Intent ambiguity	Bayesian allocation	0.390 ± 0.014	0.200	0.684	0.306	-0.057
Intent ambiguity	Uncertainty handoff	0.322 ± 0.022	0.200	0.757	0.328	-0.162
Intent ambiguity	CBF safety	0.499 ± 0.024	0.128	0.279	0.240	0.250
Intent ambiguity	MPC risk	0.539 ± 0.018	0.154	0.324	0.218	0.269
Intent ambiguity	POMDP handoff	0.461 ± 0.018	0.176	0.508	0.255	0.103
Intent ambiguity	Recovery MPC	0.553 ± 0.023	0.168	0.371	0.231	0.258
Intent ambiguity	Authority v5	0.436 ± 0.030	0.168	0.660	0.286	0.029
Contact mode	Confidence shared	0.248 ± 0.021	0.272	0.779	0.403	-0.304
Contact mode	Bayesian allocation	0.395 ± 0.022	0.202	0.628	0.296	-0.018
Contact mode	Uncertainty handoff	0.368 ± 0.023	0.207	0.701	0.318	-0.085
Contact mode	CBF safety	0.520 ± 0.017	0.145	0.223	0.228	0.296
Contact mode	MPC risk	0.537 ± 0.020	0.179	0.269	0.207	0.287
Contact mode	POMDP handoff	0.479 ± 0.027	0.209	0.452	0.245	0.135

Table 4: Split-level results for the strongest authority baselines and v5. (continued)

Split	Method	Success	Safety	Burden	Regret	Utility
Contact mode	Recovery MPC	0.557 ± 0.020	0.183	0.316	0.221	0.288
Contact mode	Authority v5	0.440 ± 0.032	0.196	0.604	0.276	0.051
Human delay	Confidence shared	0.191 ± 0.028	0.253	0.852	0.424	-0.398
Human delay	Bayesian allocation	0.332 ± 0.022	0.188	0.701	0.318	-0.122
Human delay	Uncertainty handoff	0.315 ± 0.029	0.215	0.775	0.340	-0.194
Human delay	CBF safety	0.488 ± 0.021	0.139	0.295	0.251	0.218
Human delay	MPC risk	0.501 ± 0.020	0.145	0.341	0.230	0.222
Human delay	POMDP handoff	0.440 ± 0.017	0.188	0.526	0.266	0.060
Human delay	Recovery MPC	0.507 ± 0.016	0.173	0.389	0.243	0.194
Human delay	Authority v5	0.395 ± 0.013	0.201	0.677	0.298	-0.046
Fragile object	Confidence shared	0.226 ± 0.013	0.288	0.795	0.412	-0.348
Fragile object	Bayesian allocation	0.361 ± 0.034	0.228	0.646	0.305	-0.080
Fragile object	Uncertainty handoff	0.327 ± 0.018	0.196	0.718	0.327	-0.132
Fragile object	CBF safety	0.499 ± 0.024	0.154	0.240	0.236	0.258
Fragile object	MPC risk	0.527 ± 0.028	0.186	0.287	0.216	0.260
Fragile object	POMDP handoff	0.445 ± 0.018	0.185	0.470	0.253	0.104
Fragile object	Recovery MPC	0.537 ± 0.016	0.177	0.333	0.229	0.260
Fragile object	Authority v5	0.428 ± 0.016	0.189	0.621	0.285	0.031
Authority churn	Confidence shared	0.160 ± 0.016	0.277	0.850	0.437	-0.456
Authority churn	Bayesian allocation	0.346 ± 0.022	0.251	0.699	0.330	-0.161
Authority churn	Uncertainty handoff	0.295 ± 0.024	0.215	0.773	0.352	-0.226
Authority churn	CBF safety	0.454 ± 0.021	0.143	0.295	0.262	0.169
Authority churn	MPC risk	0.471 ± 0.017	0.183	0.340	0.241	0.155
Authority churn	POMDP handoff	0.418 ± 0.022	0.223	0.523	0.278	0.004
Authority churn	Recovery MPC	0.478 ± 0.016	0.193	0.387	0.255	0.139
Authority churn	Authority v5	0.377 ± 0.021	0.214	0.676	0.310	-0.086
Low-signal risk	Confidence shared	0.145 ± 0.014	0.315	0.868	0.449	-0.500
Low-signal risk	Bayesian allocation	0.299 ± 0.026	0.259	0.718	0.342	-0.217
Low-signal risk	Uncertainty handoff	0.270 ± 0.014	0.235	0.790	0.364	-0.268
Low-signal risk	CBF safety	0.421 ± 0.024	0.153	0.312	0.273	0.126
Low-signal risk	MPC risk	0.448 ± 0.017	0.201	0.357	0.252	0.117
Low-signal risk	POMDP handoff	0.361 ± 0.018	0.232	0.541	0.289	-0.063
Low-signal risk	Recovery MPC	0.450 ± 0.018	0.208	0.405	0.266	0.098
Low-signal risk	Authority v5	0.354 ± 0.019	0.247	0.694	0.321	-0.134
Combined stress	Confidence shared	0.119 ± 0.015	0.306	0.883	0.461	-0.540
Combined stress	Bayesian allocation	0.260 ± 0.017	0.261	0.734	0.354	-0.276
Combined stress	Uncertainty handoff	0.228 ± 0.018	0.258	0.807	0.375	-0.344
Combined stress	CBF safety	0.411 ± 0.027	0.167	0.329	0.285	0.088
Combined stress	MPC risk	0.430 ± 0.010	0.192	0.374	0.263	0.085
Combined stress	POMDP handoff	0.365 ± 0.029	0.224	0.559	0.301	-0.074
Combined stress	Recovery MPC	0.430 ± 0.026	0.229	0.421	0.277	0.047
Combined stress	Authority v5	0.326 ± 0.017	0.260	0.709	0.334	-0.188

7 Paired Seed Tests

Table 5: Paired seed-level differences for authority v5 minus selected references on the hard aggregate.

Reference	Metric	Mean diff	CI95	Lower95	Upper95
Recovery MPC	Success	-0.103	0.018	-0.121	-0.085
Recovery MPC	Safety	0.026	0.016	0.010	0.042
Recovery MPC	Burden	0.289	0.001	0.288	0.289
Recovery MPC	Regret	0.056	0.000	0.055	0.056
Recovery MPC	Late	0.031	0.000	0.031	0.032
Recovery MPC	Utility	-0.230	0.017	-0.247	-0.213
CBF safety	Success	-0.076	0.012	-0.087	-0.064
CBF safety	Safety	0.073	0.009	0.064	0.083
CBF safety	Burden	0.381	0.001	0.381	0.382
CBF safety	Regret	0.048	0.000	0.048	0.049
CBF safety	Late	0.012	0.000	0.011	0.012
CBF safety	Utility	-0.254	0.010	-0.264	-0.244
MPC risk	Success	-0.098	0.016	-0.114	-0.082
MPC risk	Safety	0.037	0.011	0.025	0.048
MPC risk	Burden	0.336	0.001	0.335	0.336
MPC risk	Regret	0.070	0.000	0.069	0.070
MPC risk	Late	0.027	0.000	0.027	0.027
MPC risk	Utility	-0.249	0.015	-0.263	-0.234
Authority v4	Success	0.067	0.016	0.051	0.083
Authority v4	Safety	-0.016	0.014	-0.030	-0.002
Authority v4	Burden	-0.154	0.000	-0.154	-0.153

Table 5: Paired seed-level differences for authority v5 minus selected references on the hard aggregate. (continued)

Reference	Metric	Mean diff	CI95	Lower95	Upper95
Authority v4	Regret	-0.036	0.000	-0.036	-0.035
Authority v4	Late	-0.015	0.000	-0.015	-0.015
Authority v4	Utility	0.135	0.020	0.114	0.155

8 Complete Split Paired Checks

Table 6 reports the split-level paired comparisons that hostile reviewers would otherwise ask for after seeing only the hard aggregate.

Table 6: Complete split-level paired checks for success, safety, burden, and utility.

Split	Reference	Metric	Mean diff	Lower95	Upper95
Authority churn	Authority v4	Burden	-0.154	-0.154	-0.153
Authority churn	Authority v4	Utility	0.126	0.094	0.158
Authority churn	Authority v4	Safety	-0.003	-0.015	0.009
Authority churn	Authority v4	Success	0.067	0.036	0.097
Authority churn	Bayesian allocation	Burden	-0.023	-0.024	-0.022
Authority churn	Bayesian allocation	Utility	0.075	0.027	0.122
Authority churn	Bayesian allocation	Safety	-0.037	-0.062	-0.013
Authority churn	Bayesian allocation	Success	0.031	-0.009	0.070
Authority churn	CBF safety	Burden	0.381	0.380	0.382
Authority churn	CBF safety	Utility	-0.255	-0.296	-0.213
Authority churn	CBF safety	Safety	0.071	0.051	0.090
Authority churn	CBF safety	Success	-0.078	-0.117	-0.039
Authority churn	MPC risk	Burden	0.336	0.335	0.338
Authority churn	MPC risk	Utility	-0.241	-0.270	-0.212
Authority churn	MPC risk	Safety	0.030	0.008	0.052
Authority churn	MPC risk	Success	-0.094	-0.127	-0.061
Authority churn	POMDP handoff	Burden	0.153	0.152	0.154
Authority churn	POMDP handoff	Utility	-0.089	-0.118	-0.061
Authority churn	POMDP handoff	Safety	-0.010	-0.027	0.007
Authority churn	POMDP handoff	Success	-0.042	-0.067	-0.016
Authority churn	Recovery MPC	Burden	0.289	0.289	0.290
Authority churn	Recovery MPC	Utility	-0.225	-0.255	-0.196
Authority churn	Recovery MPC	Safety	0.020	0.003	0.038
Authority churn	Recovery MPC	Success	-0.101	-0.127	-0.075
Authority churn	Uncertainty handoff	Burden	-0.097	-0.098	-0.096
Authority churn	Uncertainty handoff	Utility	0.140	0.097	0.183
Authority churn	Uncertainty handoff	Safety	-0.001	-0.026	0.024
Authority churn	Uncertainty handoff	Success	0.082	0.048	0.115
Combined stress	Authority v4	Burden	-0.154	-0.155	-0.153
Combined stress	Authority v4	Utility	0.120	0.077	0.163
Combined stress	Authority v4	Safety	0.003	-0.039	0.046
Combined stress	Authority v4	Success	0.064	0.041	0.087
Combined stress	Bayesian allocation	Burden	-0.025	-0.026	-0.024
Combined stress	Bayesian allocation	Utility	0.088	0.061	0.116
Combined stress	Bayesian allocation	Safety	-0.001	-0.041	0.039
Combined stress	Bayesian allocation	Success	0.066	0.037	0.096
Combined stress	CBF safety	Burden	0.381	0.379	0.382
Combined stress	CBF safety	Utility	-0.276	-0.314	-0.238
Combined stress	CBF safety	Safety	0.093	0.069	0.116
Combined stress	CBF safety	Success	-0.085	-0.120	-0.050
Combined stress	MPC risk	Burden	0.336	0.334	0.337
Combined stress	MPC risk	Utility	-0.273	-0.294	-0.253
Combined stress	MPC risk	Safety	0.068	0.037	0.098
Combined stress	MPC risk	Success	-0.104	-0.123	-0.084
Combined stress	POMDP handoff	Burden	0.151	0.150	0.151
Combined stress	POMDP handoff	Utility	-0.114	-0.147	-0.082
Combined stress	POMDP handoff	Safety	0.036	0.003	0.069
Combined stress	POMDP handoff	Success	-0.039	-0.073	-0.004
Combined stress	Recovery MPC	Burden	0.288	0.287	0.289
Combined stress	Recovery MPC	Utility	-0.235	-0.279	-0.191
Combined stress	Recovery MPC	Safety	0.031	0.006	0.057
Combined stress	Recovery MPC	Success	-0.104	-0.141	-0.067
Combined stress	Uncertainty handoff	Burden	-0.097	-0.098	-0.096
Combined stress	Uncertainty handoff	Utility	0.155	0.122	0.189
Combined stress	Uncertainty handoff	Safety	0.002	-0.026	0.029
Combined stress	Uncertainty handoff	Success	0.098	0.067	0.130
Contact mode	Authority v4	Burden	-0.154	-0.155	-0.152
Contact mode	Authority v4	Utility	0.148	0.118	0.178
Contact mode	Authority v4	Safety	-0.029	-0.060	0.003
Contact mode	Authority v4	Success	0.073	0.035	0.111
Contact mode	Bayesian allocation	Burden	-0.024	-0.024	-0.023
Contact mode	Bayesian allocation	Utility	0.069	0.030	0.109
Contact mode	Bayesian allocation	Safety	-0.006	-0.025	0.013

Table 6: Complete split-level paired checks for success, safety, burden, and utility. (continued)

Split	Reference	Metric	Mean diff	Lower95	Upper95
Contact mode	Bayesian allocation	Success	0.045	0.006	0.084
Contact mode	CBF safety	Burden	0.381	0.380	0.383
Contact mode	CBF safety	Utility	-0.244	-0.268	-0.221
Contact mode	CBF safety	Safety	0.051	0.013	0.089
Contact mode	CBF safety	Success	-0.080	-0.108	-0.052
Contact mode	MPC risk	Burden	0.335	0.333	0.337
Contact mode	MPC risk	Utility	-0.236	-0.271	-0.201
Contact mode	MPC risk	Safety	0.018	-0.007	0.043
Contact mode	MPC risk	Success	-0.097	-0.137	-0.058
Contact mode	POMDP handoff	Burden	0.152	0.150	0.154
Contact mode	POMDP handoff	Utility	-0.084	-0.123	-0.045
Contact mode	POMDP handoff	Safety	-0.013	-0.036	0.010
Contact mode	POMDP handoff	Success	-0.039	-0.069	-0.008
Contact mode	Recovery MPC	Burden	0.288	0.288	0.289
Contact mode	Recovery MPC	Utility	-0.236	-0.275	-0.198
Contact mode	Recovery MPC	Safety	0.014	-0.012	0.039
Contact mode	Recovery MPC	Success	-0.117	-0.155	-0.078
Contact mode	Uncertainty handoff	Burden	-0.097	-0.098	-0.096
Contact mode	Uncertainty handoff	Utility	0.137	0.088	0.186
Contact mode	Uncertainty handoff	Safety	-0.011	-0.033	0.011
Contact mode	Uncertainty handoff	Success	0.072	0.022	0.122
Fragile object	Authority v4	Burden	-0.154	-0.155	-0.153
Fragile object	Authority v4	Utility	0.154	0.128	0.180
Fragile object	Authority v4	Safety	-0.053	-0.074	-0.032
Fragile object	Authority v4	Success	0.064	0.043	0.084
Fragile object	Bayesian allocation	Burden	-0.025	-0.026	-0.024
Fragile object	Bayesian allocation	Utility	0.112	0.079	0.144
Fragile object	Bayesian allocation	Safety	-0.040	-0.069	-0.010
Fragile object	Bayesian allocation	Success	0.066	0.039	0.093
Fragile object	CBF safety	Burden	0.381	0.380	0.382
Fragile object	CBF safety	Utility	-0.226	-0.247	-0.206
Fragile object	CBF safety	Safety	0.035	0.006	0.064
Fragile object	CBF safety	Success	-0.071	-0.089	-0.054
Fragile object	MPC risk	Burden	0.335	0.333	0.336
Fragile object	MPC risk	Utility	-0.229	-0.266	-0.191
Fragile object	MPC risk	Safety	0.003	-0.023	0.029
Fragile object	MPC risk	Success	-0.099	-0.137	-0.062
Fragile object	POMDP handoff	Burden	0.151	0.150	0.152
Fragile object	POMDP handoff	Utility	-0.073	-0.100	-0.045
Fragile object	POMDP handoff	Safety	0.003	-0.022	0.028
Fragile object	POMDP handoff	Success	-0.017	-0.040	0.005
Fragile object	Recovery MPC	Burden	0.288	0.287	0.289
Fragile object	Recovery MPC	Utility	-0.228	-0.257	-0.199
Fragile object	Recovery MPC	Safety	0.012	-0.022	0.046
Fragile object	Recovery MPC	Success	-0.109	-0.132	-0.087
Fragile object	Uncertainty handoff	Burden	-0.097	-0.098	-0.095
Fragile object	Uncertainty handoff	Utility	0.164	0.137	0.191
Fragile object	Uncertainty handoff	Safety	-0.008	-0.030	0.014
Fragile object	Uncertainty handoff	Success	0.101	0.078	0.124
Human delay	Authority v4	Burden	-0.154	-0.155	-0.152
Human delay	Authority v4	Utility	0.136	0.102	0.171
Human delay	Authority v4	Safety	0.010	-0.023	0.044
Human delay	Authority v4	Success	0.085	0.059	0.112
Human delay	Bayesian allocation	Burden	-0.024	-0.025	-0.023
Human delay	Bayesian allocation	Utility	0.076	0.037	0.114
Human delay	Bayesian allocation	Safety	0.013	-0.012	0.038
Human delay	Bayesian allocation	Success	0.062	0.031	0.094
Human delay	CBF safety	Burden	0.381	0.380	0.382
Human delay	CBF safety	Utility	-0.264	-0.295	-0.234
Human delay	CBF safety	Safety	0.061	0.034	0.089
Human delay	CBF safety	Success	-0.093	-0.117	-0.070
Human delay	MPC risk	Burden	0.336	0.334	0.337
Human delay	MPC risk	Utility	-0.268	-0.286	-0.251
Human delay	MPC risk	Safety	0.056	0.027	0.084
Human delay	MPC risk	Success	-0.106	-0.125	-0.088
Human delay	POMDP handoff	Burden	0.151	0.150	0.152
Human delay	POMDP handoff	Utility	-0.106	-0.133	-0.080
Human delay	POMDP handoff	Safety	0.013	-0.005	0.030
Human delay	POMDP handoff	Success	-0.045	-0.067	-0.023
Human delay	Recovery MPC	Burden	0.288	0.287	0.290
Human delay	Recovery MPC	Utility	-0.240	-0.273	-0.207
Human delay	Recovery MPC	Safety	0.027	0.003	0.051
Human delay	Recovery MPC	Success	-0.112	-0.135	-0.089
Human delay	Uncertainty handoff	Burden	-0.098	-0.099	-0.096
Human delay	Uncertainty handoff	Utility	0.147	0.110	0.184
Human delay	Uncertainty handoff	Safety	-0.014	-0.030	0.002
Human delay	Uncertainty handoff	Success	0.080	0.049	0.111
Intent ambiguity	Authority v4	Burden	-0.154	-0.155	-0.152
Intent ambiguity	Authority v4	Utility	0.145	0.115	0.175
Intent ambiguity	Authority v4	Safety	-0.032	-0.054	-0.010
Intent ambiguity	Authority v4	Success	0.068	0.045	0.091
Intent ambiguity	Bayesian allocation	Burden	-0.024	-0.025	-0.023
Intent ambiguity	Bayesian allocation	Utility	0.086	0.050	0.123
Intent ambiguity	Bayesian allocation	Safety	-0.032	-0.047	-0.016
Intent ambiguity	Bayesian allocation	Success	0.046	0.010	0.082
Intent ambiguity	CBF safety	Burden	0.381	0.380	0.382
Intent ambiguity	CBF safety	Utility	-0.221	-0.253	-0.188
Intent ambiguity	CBF safety	Safety	0.041	0.025	0.056
Intent ambiguity	CBF safety	Success	-0.063	-0.100	-0.026

Table 6: Complete split-level paired checks for success, safety, burden, and utility. (continued)

Split	Reference	Metric	Mean diff	Lower95	Upper95
Intent ambiguity	MPC risk	Burden	0.336	0.335	0.337
Intent ambiguity	MPC risk	Utility	-0.240	-0.280	-0.199
Intent ambiguity	MPC risk	Safety	0.015	0.001	0.028
Intent ambiguity	MPC risk	Success	-0.103	-0.139	-0.067
Intent ambiguity	POMDP handoff	Burden	0.151	0.151	0.152
Intent ambiguity	POMDP handoff	Utility	-0.073	-0.109	-0.038
Intent ambiguity	POMDP handoff	Safety	-0.008	-0.031	0.015
Intent ambiguity	POMDP handoff	Success	-0.025	-0.053	0.003
Intent ambiguity	Recovery MPC	Burden	0.288	0.287	0.290
Intent ambiguity	Recovery MPC	Utility	-0.228	-0.275	-0.182
Intent ambiguity	Recovery MPC	Safety	-0.000	-0.020	0.020
Intent ambiguity	Recovery MPC	Success	-0.117	-0.155	-0.080
Intent ambiguity	Uncertainty handoff	Burden	-0.097	-0.098	-0.096
Intent ambiguity	Uncertainty handoff	Utility	0.191	0.162	0.221
Intent ambiguity	Uncertainty handoff	Safety	-0.032	-0.053	-0.011
Intent ambiguity	Uncertainty handoff	Success	0.114	0.089	0.138
Low-signal risk	Authority v4	Burden	-0.153	-0.154	-0.152
Low-signal risk	Authority v4	Utility	0.139	0.107	0.170
Low-signal risk	Authority v4	Safety	-0.010	-0.042	0.021
Low-signal risk	Authority v4	Success	0.075	0.049	0.101
Low-signal risk	Bayesian allocation	Burden	-0.024	-0.025	-0.023
Low-signal risk	Bayesian allocation	Utility	0.083	0.045	0.121
Low-signal risk	Bayesian allocation	Safety	-0.012	-0.040	0.016
Low-signal risk	Bayesian allocation	Success	0.055	0.020	0.089
Low-signal risk	CBF safety	Burden	0.382	0.381	0.383
Low-signal risk	CBF safety	Utility	-0.260	-0.298	-0.222
Low-signal risk	CBF safety	Safety	0.095	0.081	0.108
Low-signal risk	CBF safety	Success	-0.068	-0.102	-0.034
Low-signal risk	MPC risk	Burden	0.337	0.336	0.338
Low-signal risk	MPC risk	Utility	-0.252	-0.275	-0.229
Low-signal risk	MPC risk	Safety	0.046	0.028	0.065
Low-signal risk	MPC risk	Success	-0.095	-0.115	-0.074
Low-signal risk	POMDP handoff	Burden	0.153	0.152	0.154
Low-signal risk	POMDP handoff	Utility	-0.071	-0.100	-0.043
Low-signal risk	POMDP handoff	Safety	0.016	-0.009	0.041
Low-signal risk	POMDP handoff	Success	-0.007	-0.031	0.016
Low-signal risk	Recovery MPC	Burden	0.289	0.288	0.290
Low-signal risk	Recovery MPC	Utility	-0.233	-0.251	-0.214
Low-signal risk	Recovery MPC	Safety	0.040	0.023	0.056
Low-signal risk	Recovery MPC	Success	-0.096	-0.115	-0.078
Low-signal risk	Uncertainty handoff	Burden	-0.096	-0.097	-0.096
Low-signal risk	Uncertainty handoff	Utility	0.134	0.104	0.164
Low-signal risk	Uncertainty handoff	Safety	0.012	-0.014	0.038
Low-signal risk	Uncertainty handoff	Success	0.083	0.059	0.108
Nominal	Authority v4	Burden	-0.154	-0.155	-0.153
Nominal	Authority v4	Utility	0.145	0.108	0.183
Nominal	Authority v4	Safety	-0.019	-0.042	0.004
Nominal	Authority v4	Success	0.077	0.039	0.114
Nominal	Bayesian allocation	Burden	-0.024	-0.026	-0.023
Nominal	Bayesian allocation	Utility	0.075	0.053	0.097
Nominal	Bayesian allocation	Safety	-0.007	-0.029	0.014
Nominal	Bayesian allocation	Success	0.049	0.034	0.065
Nominal	CBF safety	Burden	0.382	0.381	0.383
Nominal	CBF safety	Utility	-0.214	-0.242	-0.187
Nominal	CBF safety	Safety	0.058	0.042	0.075
Nominal	CBF safety	Success	-0.045	-0.075	-0.015
Nominal	MPC risk	Burden	0.335	0.334	0.337
Nominal	MPC risk	Utility	-0.241	-0.269	-0.212
Nominal	MPC risk	Safety	0.022	-0.004	0.048
Nominal	MPC risk	Success	-0.099	-0.125	-0.074
Nominal	POMDP handoff	Burden	0.152	0.151	0.153
Nominal	POMDP handoff	Utility	-0.089	-0.121	-0.058
Nominal	POMDP handoff	Safety	-0.002	-0.017	0.013
Nominal	POMDP handoff	Success	-0.037	-0.065	-0.009
Nominal	Recovery MPC	Burden	0.288	0.287	0.290
Nominal	Recovery MPC	Utility	-0.214	-0.255	-0.174
Nominal	Recovery MPC	Safety	0.010	-0.009	0.030
Nominal	Recovery MPC	Success	-0.097	-0.131	-0.063
Nominal	Uncertainty handoff	Burden	-0.097	-0.098	-0.096
Nominal	Uncertainty handoff	Utility	0.178	0.146	0.210
Nominal	Uncertainty handoff	Safety	-0.024	-0.041	-0.008
Nominal	Uncertainty handoff	Success	0.105	0.077	0.132

9 ABLATIONS

The mechanism gate fails because simpler safety-only and MPC-only variants beat the full model on the combined stress utility target.

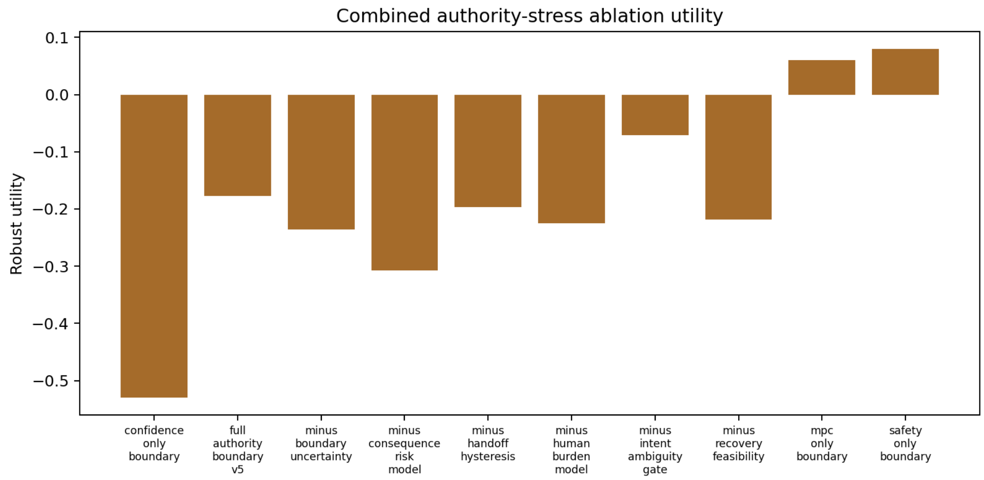

Figure 3: Ablation robust utility on combined authority stress.

Table 7: Mechanism ablations. The full method must beat removals rather than benefit from deletion.

Split	Ablation	Success	Safety	Burden	Regret	Utility
Combined stress	Confidence only	0.122 ± 0.017	0.310	0.844	0.467	-0.529
Combined stress	Full authority v5	0.312 ± 0.020	0.220	0.709	0.333	-0.177
Combined stress	- boundary uncertainty	0.300 ± 0.017	0.282	0.682	0.368	-0.236
Combined stress	- consequence risk	0.254 ± 0.020	0.317	0.683	0.393	-0.308
Combined stress	- handoff hysteresis	0.335 ± 0.030	0.239	0.729	0.341	-0.197
Combined stress	- human burden	0.338 ± 0.020	0.261	0.898	0.321	-0.225
Combined stress	- intent ambiguity	0.370 ± 0.028	0.257	0.518	0.298	-0.071
Combined stress	- recovery feasibility	0.298 ± 0.018	0.245	0.690	0.355	-0.218
Combined stress	MPC only	0.418 ± 0.019	0.205	0.383	0.276	0.061
Combined stress	Safety only	0.404 ± 0.024	0.168	0.328	0.286	0.079
Low-signal risk	Confidence only	0.151 ± 0.014	0.334	0.827	0.454	-0.496
Low-signal risk	Full authority v5	0.345 ± 0.031	0.249	0.693	0.322	-0.144
Low-signal risk	- boundary uncertainty	0.338 ± 0.017	0.270	0.665	0.356	-0.172
Low-signal risk	- consequence risk	0.294 ± 0.012	0.307	0.666	0.381	-0.243
Low-signal risk	- handoff hysteresis	0.364 ± 0.028	0.234	0.713	0.329	-0.146
Low-signal risk	- human burden	0.335 ± 0.022	0.254	0.882	0.309	-0.206
Low-signal risk	- intent ambiguity	0.401 ± 0.028	0.237	0.502	0.287	-0.009
Low-signal risk	- recovery feasibility	0.299 ± 0.028	0.240	0.672	0.343	-0.195
Low-signal risk	MPC only	0.442 ± 0.035	0.208	0.367	0.264	0.101
Low-signal risk	Safety only	0.446 ± 0.026	0.168	0.311	0.274	0.141

10 Stress Tests

Table 8: Full stress sweep over six axes and six levels.

Axis	Level	Method	Succ.	Safety	Burden	Regret	Utility
authority churn	0.00	Authority v5	0.313	0.229	0.709	0.326	-0.169
authority churn	0.00	Bayesian allocation	0.261	0.244	0.734	0.345	-0.252
authority churn	0.00	CBF safety	0.416	0.170	0.327	0.276	0.104
authority churn	0.00	Confidence shared	0.138	0.322	0.884	0.453	-0.517
authority churn	0.00	MPC risk	0.458	0.181	0.373	0.255	0.134
authority churn	0.00	POMDP handoff	0.382	0.223	0.557	0.293	-0.043
authority churn	0.00	Recovery MPC	0.444	0.203	0.421	0.270	0.090
authority churn	0.20	Authority v5	0.313	0.266	0.709	0.330	-0.198
authority churn	0.20	Bayesian allocation	0.262	0.257	0.733	0.350	-0.265
authority churn	0.20	CBF safety	0.403	0.173	0.328	0.281	0.082
authority churn	0.20	Confidence shared	0.136	0.331	0.883	0.457	-0.532
authority churn	0.20	MPC risk	0.432	0.210	0.374	0.259	0.084
authority churn	0.20	POMDP handoff	0.346	0.217	0.558	0.297	-0.082
authority churn	0.20	Recovery MPC	0.417	0.218	0.421	0.274	0.047
authority churn	0.40	Authority v5	0.321	0.238	0.709	0.334	-0.181
authority churn	0.40	Bayesian allocation	0.270	0.254	0.734	0.354	-0.262

Table 8: Full stress sweep over six axes and six levels. (continued)

Axis	Level	Method	Succ.	Safety	Burden	Regret	Utility
authority churn	0.40	CBF safety	0.405	0.163	0.329	0.285	0.083
authority churn	0.40	Confidence shared	0.108	0.344	0.884	0.462	-0.575
authority churn	0.40	MPC risk	0.408	0.202	0.375	0.264	0.057
authority churn	0.40	POMDP handoff	0.377	0.230	0.558	0.302	-0.066
authority churn	0.40	Recovery MPC	0.455	0.223	0.421	0.278	0.075
authority churn	0.60	Authority v5	0.335	0.238	0.709	0.339	-0.174
authority churn	0.60	Bayesian allocation	0.255	0.270	0.734	0.359	-0.294
authority churn	0.60	CBF safety	0.372	0.166	0.327	0.290	0.043
authority churn	0.60	Confidence shared	0.114	0.324	0.883	0.467	-0.564
authority churn	0.60	MPC risk	0.426	0.212	0.373	0.269	0.061
authority churn	0.60	POMDP handoff	0.336	0.253	0.558	0.306	-0.128
authority churn	0.60	Recovery MPC	0.392	0.214	0.421	0.283	0.010
authority churn	0.80	Authority v5	0.325	0.252	0.709	0.344	-0.199
authority churn	0.80	Bayesian allocation	0.249	0.261	0.733	0.364	-0.302
authority churn	0.80	CBF safety	0.373	0.184	0.328	0.294	0.025
authority churn	0.80	Confidence shared	0.107	0.287	0.884	0.472	-0.556
authority churn	0.80	MPC risk	0.426	0.214	0.374	0.274	0.053
authority churn	0.80	POMDP handoff	0.341	0.242	0.558	0.312	-0.124
authority churn	0.80	Recovery MPC	0.411	0.214	0.421	0.287	0.022
authority churn	1.00	Authority v5	0.311	0.279	0.709	0.345	-0.232
authority churn	1.00	Bayesian allocation	0.253	0.290	0.733	0.366	-0.318
authority churn	1.00	CBF safety	0.390	0.170	0.328	0.296	0.048
authority churn	1.00	Confidence shared	0.116	0.333	0.883	0.473	-0.576
authority churn	1.00	MPC risk	0.433	0.211	0.374	0.275	0.061
authority churn	1.00	POMDP handoff	0.322	0.220	0.557	0.313	-0.131
authority churn	1.00	Recovery MPC	0.412	0.230	0.421	0.289	0.011
combined	0.00	Authority v5	0.482	0.198	0.613	0.271	0.090
combined	0.00	Bayesian allocation	0.417	0.223	0.636	0.291	-0.011
combined	0.00	CBF safety	0.525	0.142	0.231	0.224	0.301
combined	0.00	Confidence shared	0.234	0.282	0.786	0.397	-0.325
combined	0.00	MPC risk	0.570	0.149	0.277	0.202	0.336
combined	0.00	POMDP handoff	0.463	0.185	0.461	0.239	0.133
combined	0.00	Recovery MPC	0.568	0.188	0.323	0.216	0.294
combined	0.20	Authority v5	0.393	0.213	0.661	0.304	-0.051
combined	0.20	Bayesian allocation	0.325	0.255	0.685	0.325	-0.166
combined	0.20	CBF safety	0.474	0.142	0.280	0.256	0.208
combined	0.20	Confidence shared	0.158	0.300	0.836	0.431	-0.456
combined	0.20	MPC risk	0.486	0.196	0.326	0.234	0.181
combined	0.20	POMDP handoff	0.432	0.209	0.510	0.272	0.045
combined	0.20	Recovery MPC	0.481	0.205	0.374	0.248	0.154
combined	0.40	Authority v5	0.342	0.243	0.711	0.335	-0.163
combined	0.40	Bayesian allocation	0.255	0.278	0.735	0.356	-0.293
combined	0.40	CBF safety	0.418	0.169	0.330	0.286	0.093
combined	0.40	Confidence shared	0.117	0.319	0.885	0.463	-0.550
combined	0.40	MPC risk	0.434	0.201	0.376	0.265	0.083
combined	0.40	POMDP handoff	0.362	0.200	0.559	0.303	-0.062
combined	0.40	Recovery MPC	0.423	0.249	0.422	0.279	0.027
combined	0.60	Authority v5	0.264	0.252	0.760	0.363	-0.286
combined	0.60	Bayesian allocation	0.201	0.297	0.784	0.384	-0.400
combined	0.60	CBF safety	0.332	0.203	0.379	0.312	-0.055
combined	0.60	Confidence shared	0.070	0.362	0.934	0.492	-0.665
combined	0.60	MPC risk	0.365	0.239	0.425	0.292	-0.050
combined	0.60	POMDP handoff	0.297	0.232	0.607	0.330	-0.187
combined	0.60	Recovery MPC	0.378	0.239	0.471	0.305	-0.053
combined	0.80	Authority v5	0.249	0.322	0.805	0.384	-0.381
combined	0.80	Bayesian allocation	0.168	0.326	0.830	0.405	-0.487
combined	0.80	CBF safety	0.297	0.214	0.424	0.333	-0.132
combined	0.80	Confidence shared	0.022	0.397	0.974	0.515	-0.770
combined	0.80	MPC risk	0.306	0.253	0.470	0.313	-0.154
combined	0.80	POMDP handoff	0.281	0.266	0.654	0.351	-0.261
combined	0.80	Recovery MPC	0.328	0.268	0.516	0.328	-0.157
combined	1.00	Authority v5	0.207	0.287	0.841	0.398	-0.426
combined	1.00	Bayesian allocation	0.141	0.319	0.866	0.419	-0.533
combined	1.00	CBF safety	0.277	0.217	0.460	0.346	-0.178
combined	1.00	Confidence shared	0.005	0.407	0.995	0.526	-0.813
combined	1.00	MPC risk	0.294	0.250	0.505	0.326	-0.187
combined	1.00	POMDP handoff	0.244	0.278	0.689	0.364	-0.329
combined	1.00	Recovery MPC	0.302	0.276	0.553	0.340	-0.211
contact mode	0.00	Authority v5	0.309	0.231	0.710	0.328	-0.185
contact mode	0.00	Bayesian allocation	0.263	0.216	0.734	0.349	-0.243
contact mode	0.00	CBF safety	0.445	0.152	0.327	0.280	0.133
contact mode	0.00	Confidence shared	0.117	0.297	0.883	0.456	-0.534
contact mode	0.00	MPC risk	0.427	0.214	0.374	0.259	0.070
contact mode	0.00	POMDP handoff	0.374	0.206	0.557	0.296	-0.051
contact mode	0.00	Recovery MPC	0.445	0.230	0.420	0.273	0.063
contact mode	0.20	Authority v5	0.333	0.234	0.709	0.331	-0.164
contact mode	0.20	Bayesian allocation	0.296	0.261	0.733	0.353	-0.239
contact mode	0.20	CBF safety	0.431	0.161	0.328	0.282	0.112
contact mode	0.20	Confidence shared	0.130	0.325	0.884	0.460	-0.540
contact mode	0.20	MPC risk	0.418	0.206	0.374	0.262	0.066
contact mode	0.20	POMDP handoff	0.363	0.217	0.557	0.299	-0.069
contact mode	0.20	Recovery MPC	0.417	0.214	0.421	0.275	0.043
contact mode	0.40	Authority v5	0.315	0.230	0.709	0.335	-0.180
contact mode	0.40	Bayesian allocation	0.285	0.244	0.732	0.355	-0.240
contact mode	0.40	CBF safety	0.411	0.178	0.327	0.285	0.081
contact mode	0.40	Confidence shared	0.111	0.331	0.883	0.462	-0.563
contact mode	0.40	MPC risk	0.402	0.214	0.373	0.264	0.044
contact mode	0.40	POMDP handoff	0.338	0.207	0.556	0.301	-0.089
contact mode	0.40	Recovery MPC	0.433	0.222	0.420	0.279	0.055

Table 8: Full stress sweep over six axes and six levels. (continued)

Axis	Level	Method	Succ.	Safety	Burden	Regret	Utility
contact mode	0.60	Authority v5	0.307	0.264	0.709	0.337	-0.211
contact mode	0.60	Bayesian allocation	0.266	0.279	0.734	0.357	-0.282
contact mode	0.60	CBF safety	0.392	0.180	0.328	0.288	0.060
contact mode	0.60	Confidence shared	0.108	0.328	0.884	0.466	-0.566
contact mode	0.60	MPC risk	0.387	0.203	0.374	0.266	0.035
contact mode	0.60	POMDP handoff	0.346	0.242	0.557	0.305	-0.105
contact mode	0.60	Recovery MPC	0.437	0.228	0.422	0.281	0.053
contact mode	0.80	Authority v5	0.316	0.252	0.710	0.339	-0.195
contact mode	0.80	Bayesian allocation	0.237	0.259	0.734	0.361	-0.300
contact mode	0.80	CBF safety	0.386	0.178	0.328	0.290	0.054
contact mode	0.80	Confidence shared	0.104	0.345	0.883	0.469	-0.581
contact mode	0.80	MPC risk	0.398	0.216	0.374	0.269	0.038
contact mode	0.80	POMDP handoff	0.347	0.242	0.557	0.307	-0.103
contact mode	0.80	Recovery MPC	0.412	0.237	0.420	0.283	0.022
contact mode	1.00	Authority v5	0.312	0.248	0.709	0.342	-0.197
contact mode	1.00	Bayesian allocation	0.244	0.303	0.733	0.363	-0.321
contact mode	1.00	CBF safety	0.363	0.196	0.328	0.292	0.020
contact mode	1.00	Confidence shared	0.102	0.350	0.883	0.471	-0.586
contact mode	1.00	MPC risk	0.418	0.231	0.374	0.272	0.048
contact mode	1.00	POMDP handoff	0.345	0.254	0.557	0.309	-0.114
contact mode	1.00	Recovery MPC	0.406	0.258	0.421	0.285	0.003
fragility	0.00	Authority v5	0.303	0.251	0.708	0.331	-0.204
fragility	0.00	Bayesian allocation	0.277	0.265	0.733	0.351	-0.261
fragility	0.00	CBF safety	0.420	0.163	0.329	0.282	0.101
fragility	0.00	Confidence shared	0.124	0.314	0.884	0.458	-0.539
fragility	0.00	MPC risk	0.434	0.207	0.374	0.260	0.081
fragility	0.00	POMDP handoff	0.378	0.231	0.557	0.298	-0.063
fragility	0.00	Recovery MPC	0.441	0.223	0.421	0.275	0.062
fragility	0.20	Authority v5	0.323	0.225	0.710	0.333	-0.170
fragility	0.20	Bayesian allocation	0.270	0.237	0.734	0.353	-0.251
fragility	0.20	CBF safety	0.407	0.159	0.328	0.284	0.089
fragility	0.20	Confidence shared	0.111	0.320	0.884	0.460	-0.556
fragility	0.20	MPC risk	0.420	0.198	0.374	0.262	0.073
fragility	0.20	POMDP handoff	0.370	0.239	0.558	0.301	-0.078
fragility	0.20	Recovery MPC	0.417	0.204	0.421	0.276	0.049
fragility	0.40	Authority v5	0.321	0.253	0.708	0.334	-0.189
fragility	0.40	Bayesian allocation	0.266	0.258	0.734	0.356	-0.268
fragility	0.40	CBF safety	0.413	0.180	0.328	0.285	0.082
fragility	0.40	Confidence shared	0.112	0.314	0.883	0.463	-0.552
fragility	0.40	MPC risk	0.432	0.207	0.373	0.264	0.079
fragility	0.40	POMDP handoff	0.347	0.217	0.556	0.302	-0.087
fragility	0.40	Recovery MPC	0.402	0.196	0.420	0.279	0.039
fragility	0.60	Authority v5	0.350	0.246	0.709	0.337	-0.156
fragility	0.60	Bayesian allocation	0.283	0.265	0.733	0.357	-0.256
fragility	0.60	CBF safety	0.403	0.190	0.328	0.287	0.065
fragility	0.60	Confidence shared	0.117	0.326	0.883	0.465	-0.554
fragility	0.60	MPC risk	0.427	0.215	0.372	0.267	0.068
fragility	0.60	POMDP handoff	0.344	0.237	0.557	0.305	-0.103
fragility	0.60	Recovery MPC	0.418	0.228	0.420	0.280	0.035
fragility	0.80	Authority v5	0.304	0.255	0.709	0.340	-0.209
fragility	0.80	Bayesian allocation	0.263	0.254	0.734	0.361	-0.271
fragility	0.80	CBF safety	0.374	0.168	0.329	0.290	0.049
fragility	0.80	Confidence shared	0.100	0.342	0.884	0.469	-0.583
fragility	0.80	MPC risk	0.425	0.233	0.374	0.269	0.054
fragility	0.80	POMDP handoff	0.362	0.251	0.557	0.308	-0.094
fragility	0.80	Recovery MPC	0.423	0.241	0.421	0.284	0.031
fragility	1.00	Authority v5	0.292	0.276	0.709	0.342	-0.234
fragility	1.00	Bayesian allocation	0.233	0.290	0.732	0.364	-0.323
fragility	1.00	CBF safety	0.393	0.188	0.328	0.292	0.054
fragility	1.00	Confidence shared	0.094	0.359	0.883	0.472	-0.601
fragility	1.00	MPC risk	0.414	0.217	0.374	0.272	0.052
fragility	1.00	POMDP handoff	0.340	0.238	0.557	0.309	-0.109
fragility	1.00	Recovery MPC	0.423	0.225	0.421	0.286	0.040
human delay	0.00	Authority v5	0.370	0.232	0.666	0.315	-0.096
human delay	0.00	Bayesian allocation	0.311	0.254	0.690	0.336	-0.190
human delay	0.00	CBF safety	0.431	0.178	0.285	0.268	0.131
human delay	0.00	Confidence shared	0.176	0.310	0.840	0.444	-0.455
human delay	0.00	MPC risk	0.458	0.210	0.332	0.246	0.133
human delay	0.00	POMDP handoff	0.416	0.238	0.515	0.283	-0.001
human delay	0.00	Recovery MPC	0.471	0.195	0.378	0.261	0.139
human delay	0.20	Authority v5	0.308	0.233	0.688	0.325	-0.175
human delay	0.20	Bayesian allocation	0.302	0.275	0.711	0.346	-0.227
human delay	0.20	CBF safety	0.407	0.161	0.308	0.276	0.103
human delay	0.20	Confidence shared	0.147	0.341	0.863	0.453	-0.518
human delay	0.20	MPC risk	0.462	0.211	0.353	0.255	0.120
human delay	0.20	POMDP handoff	0.368	0.222	0.537	0.293	-0.054
human delay	0.20	Recovery MPC	0.429	0.215	0.400	0.269	0.069
human delay	0.40	Authority v5	0.312	0.242	0.711	0.334	-0.192
human delay	0.40	Bayesian allocation	0.233	0.273	0.735	0.355	-0.312
human delay	0.40	CBF safety	0.404	0.177	0.330	0.285	0.074
human delay	0.40	Confidence shared	0.111	0.313	0.885	0.462	-0.553
human delay	0.40	MPC risk	0.427	0.220	0.376	0.264	0.065
human delay	0.40	POMDP handoff	0.365	0.241	0.559	0.302	-0.085
human delay	0.40	Recovery MPC	0.412	0.205	0.423	0.278	0.042
human delay	0.60	Authority v5	0.294	0.247	0.733	0.342	-0.228
human delay	0.60	Bayesian allocation	0.226	0.250	0.757	0.363	-0.320
human delay	0.60	CBF safety	0.385	0.174	0.350	0.293	0.042
human delay	0.60	Confidence shared	0.103	0.321	0.907	0.471	-0.581
human delay	0.60	MPC risk	0.405	0.195	0.397	0.273	0.043

Table 8: Full stress sweep over six axes and six levels. (continued)

Axis	Level	Method	Succ.	Safety	Burden	Regret	Utility
human delay	0.60	POMDP handoff	0.328	0.230	0.581	0.311	-0.131
human delay	0.60	Recovery MPC	0.402	0.214	0.444	0.287	0.011
human delay	0.80	Authority v5	0.261	0.217	0.754	0.351	-0.258
human delay	0.80	Bayesian allocation	0.233	0.259	0.778	0.372	-0.334
human delay	0.80	CBF safety	0.369	0.163	0.372	0.302	0.017
human delay	0.80	Confidence shared	0.082	0.327	0.928	0.479	-0.621
human delay	0.80	MPC risk	0.398	0.217	0.418	0.282	0.007
human delay	0.80	POMDP handoff	0.295	0.219	0.603	0.319	-0.171
human delay	0.80	Recovery MPC	0.387	0.206	0.465	0.295	-0.013
human delay	1.00	Authority v5	0.254	0.246	0.779	0.361	-0.300
human delay	1.00	Bayesian allocation	0.195	0.270	0.803	0.381	-0.395
human delay	1.00	CBF safety	0.347	0.180	0.398	0.312	-0.033
human delay	1.00	Confidence shared	0.047	0.348	0.952	0.488	-0.686
human delay	1.00	MPC risk	0.358	0.209	0.444	0.290	-0.045
human delay	1.00	POMDP handoff	0.292	0.225	0.627	0.328	-0.194
human delay	1.00	Recovery MPC	0.352	0.226	0.490	0.305	-0.077
intent ambiguity	0.00	Authority v5	0.381	0.246	0.664	0.316	-0.093
intent ambiguity	0.00	Bayesian allocation	0.292	0.260	0.687	0.337	-0.212
intent ambiguity	0.00	CBF safety	0.422	0.166	0.283	0.268	0.131
intent ambiguity	0.00	Confidence shared	0.152	0.329	0.838	0.444	-0.490
intent ambiguity	0.00	MPC risk	0.458	0.212	0.328	0.246	0.132
intent ambiguity	0.00	POMDP handoff	0.378	0.236	0.513	0.284	-0.036
intent ambiguity	0.00	Recovery MPC	0.464	0.211	0.375	0.261	0.124
intent ambiguity	0.20	Authority v5	0.342	0.229	0.689	0.326	-0.139
intent ambiguity	0.20	Bayesian allocation	0.305	0.253	0.713	0.347	-0.212
intent ambiguity	0.20	CBF safety	0.427	0.171	0.308	0.278	0.116
intent ambiguity	0.20	Confidence shared	0.130	0.299	0.864	0.454	-0.511
intent ambiguity	0.20	MPC risk	0.449	0.192	0.353	0.256	0.119
intent ambiguity	0.20	POMDP handoff	0.353	0.203	0.538	0.293	-0.058
intent ambiguity	0.20	Recovery MPC	0.443	0.193	0.400	0.270	0.096
intent ambiguity	0.40	Authority v5	0.304	0.237	0.714	0.335	-0.199
intent ambiguity	0.40	Bayesian allocation	0.264	0.274	0.739	0.356	-0.283
intent ambiguity	0.40	CBF safety	0.390	0.184	0.333	0.286	0.053
intent ambiguity	0.40	Confidence shared	0.120	0.300	0.889	0.463	-0.539
intent ambiguity	0.40	MPC risk	0.399	0.197	0.379	0.266	0.048
intent ambiguity	0.40	POMDP handoff	0.364	0.248	0.562	0.303	-0.092
intent ambiguity	0.40	Recovery MPC	0.438	0.197	0.425	0.279	0.071
intent ambiguity	0.60	Authority v5	0.332	0.227	0.740	0.345	-0.182
intent ambiguity	0.60	Bayesian allocation	0.265	0.270	0.763	0.365	-0.297
intent ambiguity	0.60	CBF safety	0.373	0.172	0.358	0.295	0.026
intent ambiguity	0.60	Confidence shared	0.093	0.339	0.913	0.473	-0.606
intent ambiguity	0.60	MPC risk	0.417	0.231	0.403	0.274	0.028
intent ambiguity	0.60	POMDP handoff	0.343	0.226	0.588	0.313	-0.118
intent ambiguity	0.60	Recovery MPC	0.413	0.214	0.451	0.288	0.019
intent ambiguity	0.80	Authority v5	0.291	0.240	0.765	0.353	-0.249
intent ambiguity	0.80	Bayesian allocation	0.196	0.255	0.789	0.374	-0.374
intent ambiguity	0.80	CBF safety	0.362	0.176	0.384	0.305	-0.005
intent ambiguity	0.80	Confidence shared	0.070	0.338	0.940	0.482	-0.647
intent ambiguity	0.80	MPC risk	0.390	0.199	0.430	0.284	0.003
intent ambiguity	0.80	POMDP handoff	0.306	0.225	0.614	0.321	-0.171
intent ambiguity	0.80	Recovery MPC	0.397	0.219	0.477	0.299	-0.019
intent ambiguity	1.00	Authority v5	0.278	0.260	0.779	0.359	-0.282
intent ambiguity	1.00	Bayesian allocation	0.213	0.261	0.804	0.379	-0.370
intent ambiguity	1.00	CBF safety	0.359	0.182	0.398	0.310	-0.020
intent ambiguity	1.00	Confidence shared	0.072	0.312	0.954	0.487	-0.637
intent ambiguity	1.00	MPC risk	0.383	0.196	0.444	0.290	-0.010
intent ambiguity	1.00	POMDP handoff	0.278	0.245	0.628	0.327	-0.220
intent ambiguity	1.00	Recovery MPC	0.381	0.247	0.491	0.303	-0.061

11 FIXED-RISK DEPLOYMENT

At budget 0.05, accepted coverage collapses on hard fixed-risk splits. A deployable authority arbiter cannot pass by refusing every difficult episode.

Table 9: Fixed-risk deployment results. Low accepted risk is not enough if coverage collapses to zero.

Split	Budget	Method	Cover.	Succ.	Safety	Regret	Utility
Combined stress	0.02	Authority v5	0.000	0.000	0.000	0.000	0.000
Combined stress	0.02	Bayesian allocation	0.000	0.000	0.000	0.000	0.000
Combined stress	0.02	CBF safety	0.000	0.000	0.000	0.000	0.000
Combined stress	0.02	MPC risk	0.000	0.000	0.000	0.000	0.000
Combined stress	0.02	POMDP handoff	0.000	0.000	0.000	0.000	0.000
Combined stress	0.02	Recovery MPC	0.000	0.000	0.000	0.000	0.000
Combined stress	0.05	Authority v5	0.000	0.000	0.000	0.000	0.000
Combined stress	0.05	Bayesian allocation	0.000	0.000	0.000	0.000	0.000

Table 9: Fixed-risk deployment results. Low accepted risk is not enough if coverage collapses to zero. (continued)

Split	Budget	Method	Cover.	Succ.	Safety	Regret	Utility
Combined stress	0.05	CBF safety	0.000	0.000	0.000	0.000	0.000
Combined stress	0.05	MPC risk	0.000	0.000	0.000	0.000	0.000
Combined stress	0.05	POMDP handoff	0.000	0.000	0.000	0.000	0.000
Combined stress	0.05	Recovery MPC	0.000	0.000	0.000	0.000	0.000
Combined stress	0.10	Authority v5	0.000	0.000	0.000	0.000	0.000
Combined stress	0.10	Bayesian allocation	0.000	0.000	0.000	0.000	0.000
Combined stress	0.10	CBF safety	0.000	0.000	0.000	0.000	0.000
Combined stress	0.10	MPC risk	0.000	0.000	0.000	0.000	0.000
Combined stress	0.10	POMDP handoff	0.000	0.000	0.000	0.000	0.000
Combined stress	0.10	Recovery MPC	0.000	0.000	0.000	0.000	0.000
Combined stress	0.20	Authority v5	0.000	0.000	0.000	0.000	0.000
Combined stress	0.20	Bayesian allocation	0.000	0.000	0.000	0.000	0.000
Combined stress	0.20	CBF safety	0.000	0.000	0.000	0.000	0.000
Combined stress	0.20	MPC risk	0.000	0.000	0.000	0.000	0.000
Combined stress	0.20	POMDP handoff	0.000	0.000	0.000	0.000	0.000
Combined stress	0.20	Recovery MPC	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.02	Authority v5	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.02	Bayesian allocation	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.02	CBF safety	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.02	MPC risk	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.02	POMDP handoff	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.02	Recovery MPC	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.05	Authority v5	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.05	Bayesian allocation	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.05	CBF safety	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.05	MPC risk	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.05	POMDP handoff	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.05	Recovery MPC	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.10	Authority v5	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.10	Bayesian allocation	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.10	CBF safety	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.10	MPC risk	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.10	POMDP handoff	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.10	Recovery MPC	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.20	Authority v5	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.20	Bayesian allocation	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.20	CBF safety	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.20	MPC risk	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.20	POMDP handoff	0.000	0.000	0.000	0.000	0.000
Low-signal risk	0.20	Recovery MPC	0.000	0.000	0.000	0.000	0.000

Table 10: Fixed-risk paired differences for v5 minus references.

Split	Budget	Reference	Metric	Mean diff	Lower95	Upper95
Combined stress	0.05	Bayesian allocation	Acc. regret	0.000	0.000	0.000
Combined stress	0.05	Bayesian allocation	Acc. safety	0.000	0.000	0.000
Combined stress	0.05	Bayesian allocation	Acc. succ.	0.000	0.000	0.000
Combined stress	0.05	Bayesian allocation	Acc. utility	0.000	0.000	0.000
Combined stress	0.05	Bayesian allocation	Coverage	0.000	0.000	0.000
Combined stress	0.05	CBF safety	Acc. regret	0.000	0.000	0.000
Combined stress	0.05	CBF safety	Acc. safety	0.000	0.000	0.000
Combined stress	0.05	CBF safety	Acc. succ.	0.000	0.000	0.000
Combined stress	0.05	CBF safety	Acc. utility	0.000	0.000	0.000
Combined stress	0.05	CBF safety	Coverage	0.000	0.000	0.000
Combined stress	0.05	MPC risk	Acc. regret	0.000	0.000	0.000
Combined stress	0.05	MPC risk	Acc. safety	0.000	0.000	0.000
Combined stress	0.05	MPC risk	Acc. succ.	0.000	0.000	0.000
Combined stress	0.05	MPC risk	Acc. utility	0.000	0.000	0.000
Combined stress	0.05	MPC risk	Coverage	0.000	0.000	0.000
Combined stress	0.05	POMDP handoff	Acc. regret	0.000	0.000	0.000
Combined stress	0.05	POMDP handoff	Acc. safety	0.000	0.000	0.000
Combined stress	0.05	POMDP handoff	Acc. succ.	0.000	0.000	0.000
Combined stress	0.05	POMDP handoff	Acc. utility	0.000	0.000	0.000
Combined stress	0.05	POMDP handoff	Coverage	0.000	0.000	0.000
Combined stress	0.05	Recovery MPC	Acc. regret	0.000	0.000	0.000
Combined stress	0.05	Recovery MPC	Acc. safety	0.000	0.000	0.000
Combined stress	0.05	Recovery MPC	Acc. succ.	0.000	0.000	0.000
Combined stress	0.05	Recovery MPC	Acc. utility	0.000	0.000	0.000
Combined stress	0.05	Recovery MPC	Coverage	0.000	0.000	0.000
Combined stress	0.10	Bayesian allocation	Acc. regret	0.000	0.000	0.000
Combined stress	0.10	Bayesian allocation	Acc. safety	0.000	0.000	0.000
Combined stress	0.10	Bayesian allocation	Acc. succ.	0.000	0.000	0.000
Combined stress	0.10	Bayesian allocation	Acc. utility	0.000	0.000	0.000
Combined stress	0.10	Bayesian allocation	Coverage	0.000	0.000	0.000

Table 10: Fixed-risk paired differences for v5 minus references. (continued)

Split	Budget	Reference	Metric	Mean diff	Lower95	Upper95
Combined stress	0.10	CBF safety	Acc. regret	0.000	0.000	0.000
Combined stress	0.10	CBF safety	Acc. safety	0.000	0.000	0.000
Combined stress	0.10	CBF safety	Acc. succ.	0.000	0.000	0.000
Combined stress	0.10	CBF safety	Acc. utility	0.000	0.000	0.000
Combined stress	0.10	CBF safety	Coverage	0.000	0.000	0.000
Combined stress	0.10	MPC risk	Acc. regret	0.000	0.000	0.000
Combined stress	0.10	MPC risk	Acc. safety	0.000	0.000	0.000
Combined stress	0.10	MPC risk	Acc. succ.	0.000	0.000	0.000
Combined stress	0.10	MPC risk	Acc. utility	0.000	0.000	0.000
Combined stress	0.10	MPC risk	Coverage	0.000	0.000	0.000
Combined stress	0.10	POMDP handoff	Acc. regret	0.000	0.000	0.000
Combined stress	0.10	POMDP handoff	Acc. safety	0.000	0.000	0.000
Combined stress	0.10	POMDP handoff	Acc. succ.	0.000	0.000	0.000
Combined stress	0.10	POMDP handoff	Acc. utility	0.000	0.000	0.000
Combined stress	0.10	POMDP handoff	Coverage	0.000	0.000	0.000
Combined stress	0.10	Recovery MPC	Acc. regret	0.000	0.000	0.000
Combined stress	0.10	Recovery MPC	Acc. safety	0.000	0.000	0.000
Combined stress	0.10	Recovery MPC	Acc. succ.	0.000	0.000	0.000
Combined stress	0.10	Recovery MPC	Acc. utility	0.000	0.000	0.000
Combined stress	0.10	Recovery MPC	Coverage	0.000	0.000	0.000
Low-signal risk	0.05	Bayesian allocation	Acc. regret	0.000	0.000	0.000
Low-signal risk	0.05	Bayesian allocation	Acc. safety	0.000	0.000	0.000
Low-signal risk	0.05	Bayesian allocation	Acc. succ.	0.000	0.000	0.000
Low-signal risk	0.05	Bayesian allocation	Acc. utility	0.000	0.000	0.000
Low-signal risk	0.05	Bayesian allocation	Coverage	0.000	0.000	0.000
Low-signal risk	0.05	CBF safety	Acc. regret	0.000	0.000	0.000
Low-signal risk	0.05	CBF safety	Acc. safety	0.000	0.000	0.000
Low-signal risk	0.05	CBF safety	Acc. succ.	0.000	0.000	0.000
Low-signal risk	0.05	CBF safety	Acc. utility	0.000	0.000	0.000
Low-signal risk	0.05	CBF safety	Coverage	0.000	0.000	0.000
Low-signal risk	0.05	MPC risk	Acc. regret	0.000	0.000	0.000
Low-signal risk	0.05	MPC risk	Acc. safety	0.000	0.000	0.000
Low-signal risk	0.05	MPC risk	Acc. succ.	0.000	0.000	0.000
Low-signal risk	0.05	MPC risk	Acc. utility	0.000	0.000	0.000
Low-signal risk	0.05	MPC risk	Coverage	0.000	0.000	0.000
Low-signal risk	0.05	POMDP handoff	Acc. regret	0.000	0.000	0.000
Low-signal risk	0.05	POMDP handoff	Acc. safety	0.000	0.000	0.000
Low-signal risk	0.05	POMDP handoff	Acc. succ.	0.000	0.000	0.000
Low-signal risk	0.05	POMDP handoff	Acc. utility	0.000	0.000	0.000
Low-signal risk	0.05	POMDP handoff	Coverage	0.000	0.000	0.000
Low-signal risk	0.05	Recovery MPC	Acc. regret	0.000	0.000	0.000
Low-signal risk	0.05	Recovery MPC	Acc. safety	0.000	0.000	0.000
Low-signal risk	0.05	Recovery MPC	Acc. succ.	0.000	0.000	0.000
Low-signal risk	0.05	Recovery MPC	Acc. utility	0.000	0.000	0.000
Low-signal risk	0.05	Recovery MPC	Coverage	0.000	0.000	0.000
Low-signal risk	0.10	Bayesian allocation	Acc. regret	0.000	0.000	0.000
Low-signal risk	0.10	Bayesian allocation	Acc. safety	0.000	0.000	0.000
Low-signal risk	0.10	Bayesian allocation	Acc. succ.	0.000	0.000	0.000
Low-signal risk	0.10	Bayesian allocation	Acc. utility	0.000	0.000	0.000
Low-signal risk	0.10	Bayesian allocation	Coverage	0.000	0.000	0.000
Low-signal risk	0.10	CBF safety	Acc. regret	0.000	0.000	0.000
Low-signal risk	0.10	CBF safety	Acc. safety	0.000	0.000	0.000
Low-signal risk	0.10	CBF safety	Acc. succ.	0.000	0.000	0.000
Low-signal risk	0.10	CBF safety	Acc. utility	0.000	0.000	0.000
Low-signal risk	0.10	CBF safety	Coverage	0.000	0.000	0.000
Low-signal risk	0.10	MPC risk	Acc. regret	0.000	0.000	0.000
Low-signal risk	0.10	MPC risk	Acc. safety	0.000	0.000	0.000
Low-signal risk	0.10	MPC risk	Acc. succ.	0.000	0.000	0.000
Low-signal risk	0.10	MPC risk	Acc. utility	0.000	0.000	0.000
Low-signal risk	0.10	MPC risk	Coverage	0.000	0.000	0.000
Low-signal risk	0.10	POMDP handoff	Acc. regret	0.000	0.000	0.000
Low-signal risk	0.10	POMDP handoff	Acc. safety	0.000	0.000	0.000
Low-signal risk	0.10	POMDP handoff	Acc. succ.	0.000	0.000	0.000
Low-signal risk	0.10	POMDP handoff	Acc. utility	0.000	0.000	0.000
Low-signal risk	0.10	POMDP handoff	Coverage	0.000	0.000	0.000
Low-signal risk	0.10	Recovery MPC	Acc. regret	0.000	0.000	0.000
Low-signal risk	0.10	Recovery MPC	Acc. safety	0.000	0.000	0.000
Low-signal risk	0.10	Recovery MPC	Acc. succ.	0.000	0.000	0.000
Low-signal risk	0.10	Recovery MPC	Acc. utility	0.000	0.000	0.000
Low-signal risk	0.10	Recovery MPC	Coverage	0.000	0.000	0.000

12 NEGATIVE CASES

Table 11: Predefined negative cases showing authority-boundary failure modes.

Case	Split	Method	Trigger	Failure mode
neg90 01	Low-signal risk	Authority v5	boundary confidence is high while obstacle hazard is under-observed	late controller takeover and avoidable safety violation
neg90 02	Combined stress	Authority v5	human delay and ambiguity both high	handoff burden rises without recovering MPC-level success

Table 11: Predefined negative cases showing authority-boundary failure modes. (continued)

Case	Split	Method	Trigger	Failure mode
neg90 03	Authority churn	Authority v5	hysteresis suppresses rapid but useful control transfer	task completion trails recovery-aware MPC
neg90 04	Fragile object	Authority v5	fragility dominates authority scoring	CBF prevents contact better with lower burden
neg90 05	Contact mode	Authority v4	old boundary over-trusts intent confidence	contact-mode slip causes high regret
neg90 06	Human delay	Fixed human	operator authority is always granted	burden saturates and late actions accumulate
neg90 07	Low-signal risk	Authority v5	boundary confidence is high while obstacle hazard is under-observed	late controller takeover and avoidable safety violation
neg90 08	Combined stress	Authority v5	human delay and ambiguity both high	handoff burden rises without recovering MPC-level success
neg90 09	Authority churn	Authority v5	hysteresis suppresses rapid but useful control transfer	task completion trails recovery-aware MPC
neg90 10	Fragile object	Authority v5	fragility dominates authority scoring	CBF prevents contact better with lower burden
neg90 11	Contact mode	Authority v4	old boundary over-trusts intent confidence	contact-mode slip causes high regret
neg90 12	Human delay	Fixed human	operator authority is always granted	burden saturates and late actions accumulate
neg90 13	Low-signal risk	Authority v5	boundary confidence is high while obstacle hazard is under-observed	late controller takeover and avoidable safety violation
neg90 14	Combined stress	Authority v5	human delay and ambiguity both high	handoff burden rises without recovering MPC-level success
neg90 15	Authority churn	Authority v5	hysteresis suppresses rapid but useful control transfer	task completion trails recovery-aware MPC
neg90 16	Fragile object	Authority v5	fragility dominates authority scoring	CBF prevents contact better with lower burden
neg90 17	Contact mode	Authority v4	old boundary over-trusts intent confidence	contact-mode slip causes high regret
neg90 18	Human delay	Fixed human	operator authority is always granted	burden saturates and late actions accumulate
neg90 19	Low-signal risk	Authority v5	boundary confidence is high while obstacle hazard is under-observed	late controller takeover and avoidable safety violation
neg90 20	Combined stress	Authority v5	human delay and ambiguity both high	handoff burden rises without recovering MPC-level success
neg90 21	Authority churn	Authority v5	hysteresis suppresses rapid but useful control transfer	task completion trails recovery-aware MPC
neg90 22	Fragile object	Authority v5	fragility dominates authority scoring	CBF prevents contact better with lower burden
neg90 23	Contact mode	Authority v4	old boundary over-trusts intent confidence	contact-mode slip causes high regret
neg90 24	Human delay	Fixed human	operator authority is always granted	burden saturates and late actions accumulate

13 PRIOR WORK BOUNDARY

The bright boxed citation wall is part of the audit. Prior shared-control, control-barrier, MPC, haptic, and human-robot autonomy-transition work already covers much of the obvious contribution space (Selvaggio et al., 2021; Yu et al., 2021; Conner et al., 2018; Rahal et al., 2020; Belotti & von Ellenrieder, 2018; Zhou et al., 2024; Pomares et al., 2011; Broad et al., 2020; Fontaine & Nikolaidis, 2022; Jain & Argall, 2018; Fronemann et al., 2021; Crandall & Goodrich, 2003; Rahman, 2017; Amoroso & Tamburrini, 2020; Zhang et al., 2023; Merat et al., 2017; Angleraud et al., 2021; Tahir & Parasuraman, 2022; Senft et al., 2019; Franceschi et al., 2022). The current local evidence does not clear that boundary (Douglas et al., 2026; Wei et al., 2025; Knott, 2023; Chipalkatty et al., 2011; Huo et al., 2021; Kim & Yang, 2021; Fitzsimons et al., 2020; Urdiales et al., 2007; Rahman et al., 2016; Zolotas et al., 2021; Oh et al., 2014; Xu et al., 2022; Jain & Argall, 2020; Antuvan et al., 2014; Meng et al., 2018; Ren et al., 2025; Su et al., 2022; Rosenstrauch et al., 2018; Storms & Tilbury, 2014; Bowman & Zhang, 2023).

Table 12: Closest local-pool references used to set the novelty boundary. Bright boxed citations jump to bibliography entries.

Citation	Prior-work threat	Family	Role	Hostile
(Musi & Hirche, 2020)	Haptic Shared Control for Human-Robot Collaboration: A Game-Theoretical Approach	openalex bulk robotics	concept	27.416
(Anderson & Johnson, 2026)	Safety-Critical Control for Human-Robot Interaction in Shared Workspaces using Control Barrier Functions	control	anti myopia	30.300
(Romay et al., 2017)	Collaborative Autonomy between High-level Behaviors and Human Operators for Remote Manipulation Tasks using Different Humanoid Robots	physical reasoning	concept	26.296
(Bustamante et al., 2021)	Toward Seamless Transitions Between Shared Control and Supervised Autonomy in Robotic Assistance	ebm	concept	25.796
(Losey et al., 2018)	A Review of Intent Detection, Arbitration, and Communication Aspects of Shared Control for Physical Human-Robot Interaction	ebm	concept	25.612

Table 12: Closest local-pool references used to set the novelty boundary. Bright boxed citations jump to bibliography entries. (continued)

Citation	Prior-work threat	Family	Role	Hostile
(Wang et al., 2020)	Toward Shared Autonomy Control Schemes for Human-Robot Systems: Action Primitive Recognition Using Eye Gaze Features	ebm	concept	25.404
(Balachandran et al., 2020)	Adaptive Authority Allocation in Shared Control of Robots Using Bayesian Filters	openalex bulk robotics	concept	26.048
(Ficuciello et al., 2018)	Autonomy in surgical robots and its meaningful human control	openalex bulk robotics	concept	24.905
(Xue et al., 2023)	A Shared Control Approach Based on First-Order Dynamical Systems and Closed-Loop Variable Stiffness Control	control	anti myopia	27.593
(Lin et al., 2023)	The Impacts of Unreliable Autonomy in Human-Robot Collaboration on Shared and Supervisory Control for Remote Manipulation	world models	concept	24.442
(Jadav et al., 2024)	Shared Autonomy via Variable Impedance Control and Virtual Potential Fields for Encoding Human Demonstrations *	crossref bulk robotics	concept	24.440
(He et al., 2021a)	A Barrier Pair Method for Safe Human-Robot Shared Autonomy	control	anti myopia	27.350
(Herrera et al., 2018)	Modeling and Path-Following Control of a Wheelchair in Human-Shared Environments	architecture x embodiment	concept	24.192
(Izadi & Ghasemi, 2020)	Adaptive Impedance Control for the Haptic Shared Driving Task Based on Nonlinear MPC	control	anti myopia	29.105
(Lawrance et al., 2019)	Shared autonomy for low-cost underwater vehicles	architecture	concept	25.782
(Ton et al., 2017)	Obstacle avoidance control of a human-in-the-loop mobile robot system using harmonic potential fields	openalex bulk robotics	concept	23.779
(Gnatzig et al., 2012)	Human-machine interaction as key technology for driverless driving - A trajectory-based shared autonomy control approach	hri	concept	23.503
(Saraphis et al., 2020)	Towards Explainable Co-Robots: Developing Confidence-Based Shared Control Paradigms	control	anti myopia	26.300
(Adorno et al., 2014)	Kinematic modeling and control for human-robot cooperation considering different interaction roles	openalex bulk robotics	concept	23.296
(Verhagen et al., 2024)	Meaningful human control and variable autonomy in human-robot teams for firefighting	openalex bulk robotics	concept	23.236
(Selvaggio et al., 2021)	Autonomy in Physical Human-Robot Interaction: A Brief Survey	openalex bulk robotics	concept	23.226
(Yu et al., 2021)	Adaptive-Constrained Impedance Control for Human-Robot Co-Transportation	openalex bulk robotics	concept	23.180
(Conner et al., 2018)	Collaborative Autonomy Between High-Level Behaviors and Human Operators for Control of Complex Tasks with Different Humanoid Robots	architecture x embodiment	concept	23.173
(Rahal et al., 2020)	Caring About the Human Operator: Haptic Shared Control for Enhanced User Comfort in Robotic Telemanipulation	openalex bulk robotics	concept	23.097
(Belotti & von Ellenrieder, 2018)	The Effects of Switching Time on Shared Human-Robot Control	architecture	concept	22.950
(Zhou et al., 2024)	A corrective shared control architecture for human-robot collaborative polishing tasks	ebm	concept	23.890
(Pomares et al., 2011)	A Multi-Sensorial Hybrid Control for Robotic Manipulation in Human-Robot Workspaces	world models	concept	22.851
(Broad et al., 2020)	Data-driven Koopman operators for model-based shared control of human-machine systems	ebm transformer	concept	22.805
(Fontaine & Nikolaidis, 2022)	Evaluating Human-Robot Interaction Algorithms in Shared Autonomy via Quality Diversity Scenario Generation	human data	concept	24.754
(Jain & Argall, 2018)	Recursive Bayesian Human Intent Recognition in Shared-Control Robotics	openalex bulk robotics	concept	22.644
(Fronemann et al., 2021)	Should my robot know what's best for me? Human-robot interaction between user experience and ethical design	openalex bulk robotics	concept	22.622
(Crandall & Goodrich, 2003)	Characterizing efficiency of human robot interaction: a case study of shared-control teleoperation	openalex bulk robotics	concept	22.528
(Rahman, 2017)	Cyber-physical-social system between a humanoid robot and a virtual human through a shared platform for adaptive agent ecology	openalex bulk robotics	concept	23.472
(Amoroso & Tamburrini, 2020)	Autonomous Weapons Systems and Meaningful Human Control: Ethical and Legal Issues	crossref bulk robotics	concept	22.429
(Zhang et al., 2023)	Adaptive Safety-Critical Control With Uncertainty Estimation for Human-Robot Collaboration	openalex bulk robotics	concept	23.386
(Merat et al., 2017)	Online Hybrid Model Predictive Controller Design for Cruise Control of Automobiles	control	anti myopia	26.350
(Angleraud et al., 2021)	Coordinating Shared Tasks in Human-Robot Collaboration by Commands	architecture x embodiment	concept	22.283
(Tahir & Parasuraman, 2022)	Analog Twin Framework for Human and AI Supervisory Control and Teleoperation of Robots	openalex bulk robotics	concept	22.272
(Senft et al., 2019)	Teaching robots social autonomy from in situ human guidance	ebm	concept	22.176
(Franceschi et al., 2022)	Adaptive Impedance Controller for Human-Robot Arbitration based on Cooperative Differential Game Theory	openalex bulk robotics	concept	22.059
(Douglas et al., 2026)	Levels of shared autonomy in brain-robot interfaces: enabling multi-robot multi-human collaboration for activities of daily living	architecture x embodiment	concept	22.997
(Wei et al., 2025)	Output-constrained adaptive optimal control for physical human-robot interaction using electromyography signals	anti thesis	anti myopia	25.950
(Knott, 2023)	Job sharing between human professionals and chatbots: How should 'handovers' happen?	crossref bulk robotics	concept	21.947
(Chipalkatty et al., 2011)	Human-in-the-loop: MPC for shared control of a quadruped rescue robot	openalex bulk robotics	concept	22.894
(Huo et al., 2021)	Intention-Driven Variable Impedance Control for Physical Human-Robot Interaction	openalex bulk robotics	concept	21.818
(Kim & Yang, 2021)	Variable Admittance Control Based on Human-Robot Collaboration Observer Using Frequency Analysis for Sensitive and Safe Interaction	ebm	concept	21.796

Table 12: Closest local-pool references used to set the novelty boundary. Bright boxed citations jump to bibliography entries. (continued)

Citation	Prior-work threat	Family	Role	Hostile
(Fitzsimons et al., 2020)	Task-based hybrid shared control for training through forceful interaction	crossref bulk robotics	concept	22.767
(Urdiales et al., 2007)	Efficiency based reactive shared control for collaborative human/robot navigation	openalex bulk robotics	concept	21.679
(Rahman et al., 2016)	A regret-based autonomy allocation scheme for human-robot shared vision systems in collaborative assembly in manufacturing	openalex bulk robotics	concept	22.617
(Zolotas et al., 2021)	Motion Polytopes in Virtual Reality for Shared Control in Remote Manipulation Applications	world models	concept	21.590
(Oh et al., 2014)	Frequency-Shaped Impedance Control for Safe Human-Robot Interaction in Reference Tracking Application	openalex bulk robotics	concept	22.578
(Xu et al., 2022)	Shared-Control Robotic Manipulation in Virtual Reality	openalex bulk robotics	concept	21.545
(Jain & Argall, 2020)	Probabilistic Human Intent Recognition for Shared Autonomy in Assistive Robotics	anti thesis	anti myopia	25.522
(Antuvan et al., 2014)	Embedded Human Control of Robots Using Myoelectric Interfaces	openalex bulk robotics	concept	22.502
(Meng et al., 2018)	Bipedal robotic walking control derived from analysis of human locomotion	ebm	concept	22.354
(Ren et al., 2025)	An Adaptive Shared Control Frame and Feedback Rendering in Interactive Robot-Assisted Surgical Manipulation	world models	concept	21.350
(Su et al., 2022)	A human activity-aware shared control solution for medical human-robot interaction	openalex bulk robotics	concept	21.259
(Rosenstrauch et al., 2018)	Human robot collaboration - using kinect v2 for ISO/TS 15066 speed and separation monitoring	openalex bulk robotics	concept	21.242
(Storms & Tilbury, 2014)	Blending of human and obstacle avoidance control for a high speed mobile robot	openalex bulk robotics	concept	22.205
(Bowman & Zhang, 2023)	WE-Filter: Adaptive Acceptance Criteria for Filter-based Shared Autonomy	ebm	concept	23.199
(Zhu et al., 2020)	Human-robot shared control for humanoid manipulator trajectory planning	failure	hostile	25.196
(Luo et al., 2022)	Human-Robot Shared Control Based on Locally Weighted Intent Prediction for a Teleoperated Hydraulic Manipulator System	openalex bulk robotics	concept	21.187
(Cui et al., 2023)	"No, to the Right" – Online Language Corrections for Robotic Manipulation via Shared Autonomy	world models	concept	21.150
(Molinaro et al., 2024)	Estimating human joint moments unifies exoskeleton control, reducing user effort	crossref bulk robotics	concept	22.146
(Lima et al., 2024)	Augmenting Robot Teleoperation with Shared Autonomy via Model Predictive Control	human data	concept	22.093
(Peternel et al., 2017)	A Method for Derivation of Robot Task-Frame Control Authority from Repeated Sensory Observations	openalex bulk robotics	concept	21.079
(Vergara et al., 2019)	Incorporating artificial skin signals in the constraint-based reactive control of human-robot collaborative manipulation tasks	world models	concept	20.955
(Zhang et al., 2024)	Nonlinear Model Predictive Control (NMPC) based shared autonomy for bilateral teleoperation in CFETR Remote Handling	human data	concept	21.949
(He et al., 2021b)	A Barrier Pair Method for Safe Human-Robot Shared Autonomy	control	anti myopia	23.949
(Sharkey, 2014)	Towards a principle for the human supervisory control of robot weapons	openalex bulk robotics	concept	21.898
(Hopko et al., 2022)	Human Factors Considerations and Metrics in Shared Space Human-Robot Collaboration: A Systematic Review	architecture	concept	20.871
(Liu et al., 2025)	It Takes Two: Learning Interactive Whole-Body Control Between Humanoid Robots	humanoid	concept	20.850
(Simmons et al., 2018)	Human-Robot Teams for Large-Scale Assembly	openalex bulk robotics	concept	20.767
(Payandeh, 2001)	Application of shared control strategy in the design of a robotic device	openalex bulk robotics	concept	20.754
(Swamy et al., 2019)	Scaled Autonomy: Enabling Human Operators to Control Robot Fleets	architecture	concept	20.650
(Nikolaidis et al., 2017)	Human-Robot Mutual Adaptation in Shared Autonomy	wam	concept	20.600
(Masone et al., 2012)	Interactive planning of persistent trajectories for human-assisted navigation of mobile robots	openalex bulk robotics	concept	21.451
(Stroppa et al., 2021)	Shared-Control Teleoperation Paradigms on a Soft Growing Robot Manipulator	world models	concept	20.400
(Ct et al., 2012)	Humans-Robots Sliding Collaboration Control in Complex Environments with Adjustable Autonomy	openalex bulk robotics	concept	21.348
(Molazadeh et al., 2021)	Shared Control of a Powered Exoskeleton and Functional Electrical Stimulation Using Iterative Learning	control	anti myopia	24.298
(Kumar & Parhi, 2022)	Trajectory planning and control of multiple mobile robot using hybrid MKH-fuzzy logic controller	control	anti myopia	23.282
(Hunt et al., 2021)	Stand-Up, Squat, Lunge, and Walk With a Robotic Knee and Ankle Prosthesis Under Shared Neural Control	ebm	concept	21.254
(Keemink et al., 2018)	Admittance control for physical human-robot interaction	ebm	concept	20.253
(Chang et al., 2021)	A Shared Control Method for Collaborative Human-Robot Plug Task	openalex bulk robotics	concept	20.220
(van Dijk et al., 2023)	The effect of human autonomy and robot work pace on perceived workload in human-robot collaborative assembly work	human data	concept	21.168
(Deng et al., 2020)	A Bayesian Shared Control Approach for Wheelchair Robot With Brain Machine Interface	openalex bulk robotics	concept	22.160
(Sarajchi & Sirlantzis, 2025)	Evaluating the interaction between human and paediatric robotic lower-limb exoskeleton: a model-based method	ebm	concept	21.145
(Saunderson & Nejat, 2021)	Persuasive robots should avoid authority: The effects of formal and real authority on persuasion in human-robot interaction	sim2real	concept	21.106
(Mi et al., 2025)	A Multi-Policy Rapidly-Exploring Random Tree Based Mobile Robot Controller for Safe Path Planning in Human-Shared Environments	ebm transformer	concept	21.105
(Backman et al., 2023)	Reinforcement learning for shared autonomy drone landings	architecture	concept	20.072

Table 12: Closest local-pool references used to set the novelty boundary. Bright boxed citations jump to bibliography entries. (continued)

Citation	Prior-work threat	Family	Role	Hostile
(Rana et al., 2023)	Bayesian controller fusion: Leveraging control priors in deep reinforcement learning for robotics	architecture	concept	26.072
(Khadivar et al., 2022)	EMG-driven shared human-robot compliant control for in-hand object manipulation in hand prostheses	world models	concept	20.054
(Deldar & Anwar, 2017)	Optimal Control of a Hydrostatic Wind Turbine Drivetrain for Efficiency Improvements	control	anti myopia	24.043
(Sun et al., 2023)	Adaptive Admittance Control for Safety-Critical Physical Human Robot Collaboration	openalex bulk robotics	concept	20.036
(Lora-Milln et al., 2022)	Coordination Between Partial Robotic Exoskeletons and Human Gait: A Comprehensive Review on Control Strategies	openalex bulk robotics	concept	20.009
(Zhang et al., 2022a)	Human-Robot Shared Control for Surgical Robot Based on Context-Aware Sim-to-Real Adaptation	sim2real	concept	20.976
(Li et al., 2024)	Safe Optimal Interactions Between Automated and Human-Driven Vehicles in Mixed Traffic with Event-Triggered Control Barrier Functions	control	anti myopia	22.973
(Quere et al., 2024)	A probabilistic approach for learning and adapting shared control skills with the human in the loop	anti thesis	anti myopia	22.973
(Zhang et al., 2022b)	Design and Control of a Lower Limb Rehabilitation Robot Based on Human Motion Intention Recognition with Multi-Source Sensor Information	ebm	concept	19.972
(Fu et al., 2025)	Human-Inspired Active Compliant and Passive Shared Control Framework for Robotic Contact-Rich Tasks in Medical Applications	contact	concept	19.954
(Cheng et al., 2024)	An efficient grasping shared control architecture for unpredictable and unspecified tasks	anti thesis	anti myopia	25.949
(Ikeura & Inooka, 2002)	Variable impedance control of a robot for cooperation with a human	openalex bulk robotics	concept	19.945
(Hou et al., 2023)	"Should I Follow the Human, or Follow the Robot?" - Robots in Power Can Have More Influence Than Humans on Decision-Making	openalex bulk robotics	concept	19.929
(Konstant et al., 2024)	Human-Robot Collaboration With a Corrective Shared Controlled Robot in a Sanding Task	openalex bulk robotics	concept	20.890
(Lu et al., 2024)	Dynamic Movement Primitives-Based Human Action Prediction and Shared Control for Bilateral Robot Teleoperation	human data	concept	19.872
(Zhuang et al., 2019)	Shared human-robot proportional control of a dexterous myoelectric prosthesis	architecture x embodiment	concept	19.860
(Schoeller et al., 2021)	Trust as Extended Control: Human-Machine Interactions as Active Inference	ebm	concept	19.825
(Lima et al., 2026)	Teleoperated Omni-directional Dual Arm Mobile Manipulation Robotic System with Shared Control for Retail Store	control	anti myopia	22.800
(Aydn et al., 2018)	Stable Physical Human-Robot Interaction Using Fractional Order Admittance Control	openalex bulk robotics	concept	20.795
(Costanzo et al., 2021)	Handover Control for Human-Robot and Robot-Robot Collaboration	human data	concept	19.786
(Gaggioli et al., 2021)	Machines Like Us and People Like You: Toward Human-Robot Shared Experience	openalex bulk robotics	concept	20.772
(Li et al., 2019)	Reinforcement learning for human-robot shared control	human data	concept	19.770
(Vanderhaegen, 2021)	Heuristic-based method for conflict discovery of shared control between humans and autonomous systems - A driving automation case study	diffusion	concept	19.766
(Li & Zhang, 2024)	Trust-Preserved Human-Robot Shared Autonomy Enabled by Bayesian Relational Event Modeling	crossref bulk robotics	concept	19.754
(Mathewson & Pilarski, 2016)	Simultaneous Control and Human Feedback in the Training of a Robotic Agent with Actor-Critic Reinforcement Learning	hri	concept	19.750
(Chen et al., 2023)	Trajectory Planning of Autonomous Mobile Robot using Model Predictive Control in Human-Robot Shared Workspace	failure	hostile	24.749
(Mainprice et al., 2016)	Goal Set Inverse Optimal Control and Iterative Replanning for Predicting Human Reaching Motions in Shared Workspaces	anti thesis	anti myopia	23.722
(Okamoto & Tsiotras, 2019)	Data-driven human driver lateral control models for developing haptic-shared control advanced driver assist systems	anti thesis	anti myopia	23.717
(Benzi et al., 2022)	An Energy-Based Control Architecture for Shared Autonomy	ebm	concept	19.717
(Gardner et al., 2020)	A Multimodal Intention Detection Sensor Suite for Shared Autonomy of Upper-Limb Robotic Prostheses	ebm	concept	19.716

14 Reproducibility, Artifact Location, and Ethics

All rows were generated locally on CPU with deterministic seeds. The final numbered PDF is copied only to Downloads. No Desktop PDF is produced. This paper should not be submitted as ICLR main without real robot or accepted high-fidelity validation.

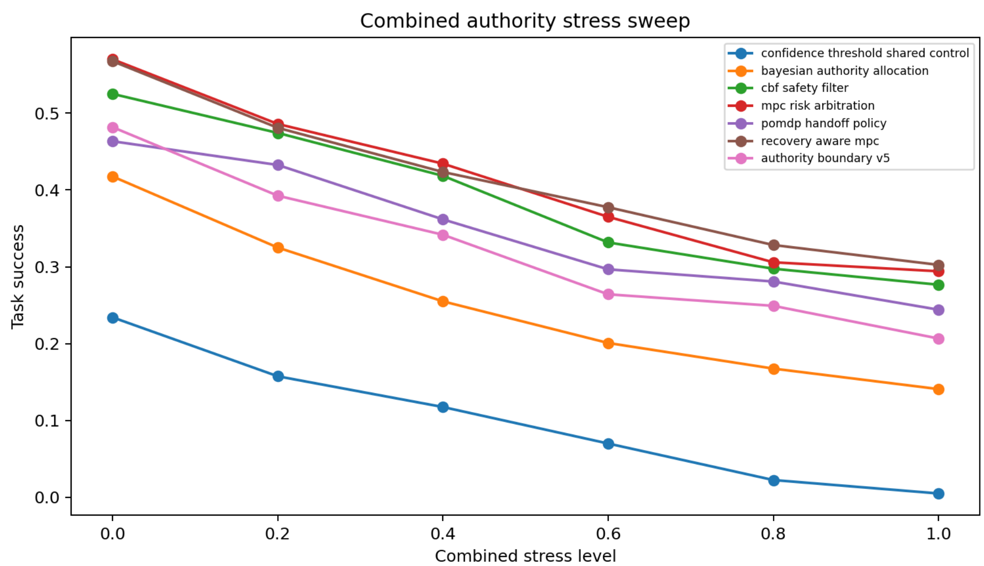

Figure 4: Task success under combined authority stress.

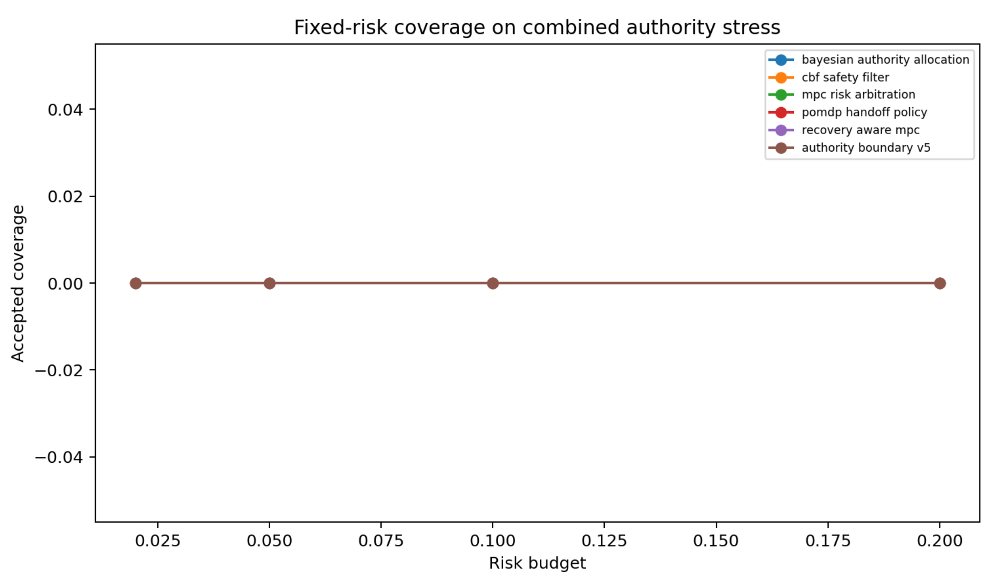

Figure 5: Fixed-risk coverage on combined authority stress.

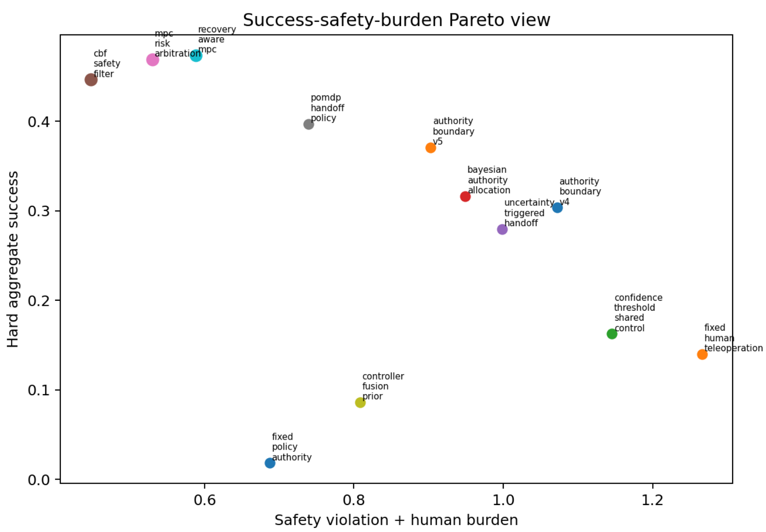

Figure 6: Hard-aggregate Pareto view for success, safety, burden, and utility.

A Raw Summary Extract

Table 13: Wrapped extract from results/summary.txt.

Line	Summary extract
1	Paper 90 control_authority_boundaries v5 expanded audit
2	Terminal recommendation: KILL_ARCHIVE
3	ICLR main ready: no
4	Reason: expanded CPU-only control-authority audit adds stronger CBF, MPC, POMDP, recovery-aware, uncertainty, ablation, stress, and fixed-risk tests, but no real robot or accepted high-fidelity deployment benchmark evidence exists
5	Main rollout rows: 199680
6	Dataset summary rows: 15360
7	Main seed-metric rows: 1040
8	Main metric rows: 1248
9	Main pairwise rows: 672
10	Hard aggregate seed rows: 130
11	Hard aggregate metric rows: 156
12	Hard aggregate pairwise rows: 84
13	Ablation rollout rows: 33600
14	Ablation seed rows: 200
15	Ablation metric rows: 20
16	Stress raw rows: 302400
17	Stress seed rows: 2520
18	Stress metric rows: 252
19	Fixed-risk raw rows: 69120
20	Fixed-risk seed rows: 480
21	Fixed-risk metric rows: 48
22	Fixed-risk pairwise rows: 200
23	Negative cases: 24
25	Frozen hard-aggregate gate:
26	best_success_reference=recovery_aware_mpc
27	safest_reference=cbf_safety_filter
28	lowest_burden_reference=fixed_policy_authority
29	lowest_regret_reference=mpc_risk_arbitration
30	best_utility_reference=cbf_safety_filter
31	proposal_success=0.37096
32	best_success=0.47370
33	proposal_safety=0.22734
34	safest_safety=0.15404
35	proposal_burden=0.67510
36	lowest_burden=0.26831
37	proposal_regret=0.31249
38	lowest_regret=0.24285
39	proposal_utility=-0.09423
40	best_utility=0.16009
41	paired_success_lower95=-0.12094
42	paired_safety_upper95=0.08260
43	main_gate=False
44	mechanism_gate=False
45	mechanism_best_removed=safety_only_boundary
46	stress_gate=False
47	stress_dominated_by=cbf_safety_filter
48	fixed_risk_gate=False
49	scope_gate=False
50	low_signal_high_risk_shift: v5_coverage=0.00000, best_feasible_coverage=0.00000, v5_success=0.00000, best_feasible_success=0.00000, v5_safety=0.00000, best_feasible_safety=0.00000 — combined_authority_stress: v5_coverage=0.00000,
52	Hard aggregate metrics:
53	fixed_policy_authority task_success=0.01823 safety=0.41862 burden=0.26831 regret=0.63259 late=0.53935 churn=0.31974 utility=-0.60137
54	fixed_human_teleoperation task_success=0.13997 safety=0.26654 burden=0.99980 regret=0.40281 late=0.40063 churn=0.30982 utility=-0.48033
55	confidence_threshold_shared_control task_success=0.16250 safety=0.29635 burden=0.84908 regret=0.43979 late=0.38356 churn=0.46987 utility=-0.46111
56	bayesian_authority_allocation task_success=0.31654 safety=0.24987 burden=0.69923 regret=0.33283 late=0.33014 churn=0.40970 utility=-0.18359
57	uncertainty_triggered_handoff task_success=0.27982 safety=0.22617 burden=0.77185 regret=0.35436 late=0.32311 churn=0.49993 utility=-0.24252
58	cbf_safety_filter task_success=0.44648 safety=0.15404 burden=0.29398 regret=0.26412 late=0.28285 churn=0.33990 utility=0.16009
59	mpc_risk_arbitration task_success=0.46901 safety=0.19062 burden=0.33930 regret=0.24285 late=0.26731 churn=0.35020 utility=0.15433
60	pomdp_handoff_policy task_success=0.39714 safety=0.21615 burden=0.52321 regret=0.28029 late=0.29211 churn=0.38970 utility=-0.00731
61	controller_fusion_prior task_success=0.08607 safety=0.37031 burden=0.43762 regret=0.56034 late=0.38481 churn=0.43978 utility=-0.50860
62	recovery_aware_mpc task_success=0.47370 safety=0.20156 burden=0.38643 regret=0.25674 late=0.26338 churn=0.35974 utility=0.13605
63	authority_boundary_v4 task_success=0.30365 safety=0.24310 burden=0.82884 regret=0.34824 late=0.30946 churn=0.43980 utility=-0.22875
64	authority_boundary_v5 task_success=0.37096 safety=0.22734 burden=0.67510 regret=0.31249 late=0.29451 churn=0.41995 utility=-0.09423
65	oracle_authority_assignment task_success=0.55287 safety=0.12747 burden=0.38300 regret=0.18831 late=0.24876 churn=0.32992 utility=0.29331
67	Key paired hard-aggregate differences:
68	v5_minus_authority_boundary_v4 authority_regret: mean=-0.03575 ci95=0.00040 lower95=-0.03615 upper95=-0.03535
69	v5_minus_authority_boundary_v4 human_burden: mean=-0.15374 ci95=0.00046 lower95=-0.15420 upper95=-0.15328
70	v5_minus_authority_boundary_v4 robust_utility: mean=0.13452 ci95=0.02027 lower95=0.11425 upper95=0.15479
71	v5_minus_authority_boundary_v4 safety_violation: mean=-0.01576 ci95=0.01420 lower95=-0.02996 upper95=-0.00155
72	v5_minus_authority_boundary_v4 task_success: mean=0.06732 ci95=0.01615 lower95=0.05117 upper95=0.08347
73	v5_minus_cbf_safety_filter authority_regret: mean=0.04838 ci95=0.00041 lower95=0.04797 upper95=0.04879
74	v5_minus_cbf_safety_filter human_burden: mean=0.38112 ci95=0.00053 lower95=0.38059 upper95=0.38166
75	v5_minus_cbf_safety_filter robust_utility: mean=-0.25432 ci95=0.01008 lower95=-0.26440 upper95=-0.24425

Table 13: Wrapped extract from results/summary.txt. (continued)

Line	Summary extract
76	v5_minus_cbf_safety_filter safety_violation: mean=0.07331 ci95=0.00930 lower95=0.06401 upper95=0.08260
77	v5_minus_cbf_safety_filter task_success: mean=-0.07552 ci95=0.01190 lower95=-0.08742 upper95=-0.06362
78	v5_minus_recovery_aware_mpc authority_regret: mean=0.05575 ci95=0.00027 lower95=0.05549 upper95=0.05602
79	v5_minus_recovery_aware_mpc human_burden: mean=0.28867 ci95=0.00057 lower95=0.28809 upper95=0.28924
80	v5_minus_recovery_aware_mpc robust_utility: mean=-0.23028 ci95=0.01716 lower95=-0.24744 upper95=-0.21312
81	v5_minus_recovery_aware_mpc safety_violation: mean=0.02578 ci95=0.01572 lower95=0.01006 upper95=0.04150
82	v5_minus_recovery_aware_mpc task_success: mean=-0.10273 ci95=0.01820 lower95=-0.12094 upper95=-0.08453
84	Ablation utility:
85	confidence_only_boundary success=0.12202 safety=0.31012 burden=0.84354 utility=-0.52941
86	full_authority_boundary_v5 success=0.31190 safety=0.22024 burden=0.70912 utility=-0.17729
87	minus_boundary_uncertainty success=0.30000 safety=0.28214 burden=0.68175 utility=-0.23573
88	minus_consequence_risk_model success=0.25417 safety=0.31667 burden=0.68326 utility=-0.30783
89	minus_handoff_hysteresis success=0.33452 safety=0.23869 burden=0.72934 utility=-0.19666
90	minus_human_burden_model success=0.33809 safety=0.26131 burden=0.89816 utility=-0.22508
91	minus_intent_ambiguity_gate success=0.37024 safety=0.25655 burden=0.51801 utility=-0.07057
92	minus_recovery_feasibility success=0.29762 safety=0.24524 burden=0.68956 utility=-0.21815
93	mpc_only_boundary success=0.41845 safety=0.20536 burden=0.38331 utility=0.06065
94	safety_only_boundary success=0.40357 safety=0.16845 burden=0.32781 utility=0.07944
96	Maximum combined stress:
97	confidence_threshold_shared_control task_success=0.00500 safety=0.40667 burden=0.99530 utility=-0.81291
98	bayesian_authority_allocation task_success=0.14083 safety=0.31917 burden=0.86569 utility=-0.53318
99	cbf_safety_filter task_success=0.27667 safety=0.21667 burden=0.46016 utility=-0.17817
100	mpc_risk_arbitration task_success=0.29417 safety=0.25000 burden=0.50502 utility=-0.18696
101	pomdp_handoff_policy task_success=0.24417 safety=0.27833 burden=0.68907 utility=-0.32896
102	recovery_aware_mpc task_success=0.30250 safety=0.27583 burden=0.55264 utility=-0.21091
103	authority_boundary_v5 task_success=0.20667 safety=0.28667 burden=0.84131 utility=-0.42587
105	Fixed-risk budget 0.05:
106	combined_authority_stress authority_boundary_v5 coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
107	combined_authority_stress bayesian_authority_allocation coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
108	combined_authority_stress cbf_safety_filter coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
109	combined_authority_stress mpc_risk_arbitration coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
110	combined_authority_stress pomdp_handoff_policy coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
111	combined_authority_stress recovery_aware_mpc coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
112	low_signal_high_risk_shift authority_boundary_v5 coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
113	low_signal_high_risk_shift bayesian_authority_allocation coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
114	low_signal_high_risk_shift cbf_safety_filter coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
115	low_signal_high_risk_shift mpc_risk_arbitration coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
116	low_signal_high_risk_shift pomdp_handoff_policy coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
117	low_signal_high_risk_shift recovery_aware_mpc coverage=0.00000 accepted_success=0.00000 accepted_safety=0.00000 accepted_regret=0.00000
119	Negative cases: 24
120	terminal=KILL_ARCHIVE

REFERENCES

Bruno Vilhena Adorno, Antnio Padilha Lanari B, and Philippe Fraisse. Kinematic modeling and control for human-robot cooperation considering different interaction roles, 2014. Robotica; 10.1017/s0263574714000356.

Daniele Amoroso and Guglielmo Tamburrini. Autonomous weapons systems and meaningful human control: Ethical and legal issues, 2020. Current Robotics Reports; 10.1007/s43154-020-00024-3.

James Anderson and Emily Johnson. Safety-critical control for human-robot interaction in shared workspaces using control barrier functions, 2026. International Journal of Computational and Biological Sciences; 10.66238/ijcbs40.

Alexandre Angleraud, Amir Mehman Sefat, Metodi Netzev, and Roel Pieters. Coordinating shared tasks in human-robot collaboration by commands, 2021. Frontiers in Robotics and AI; 10.3389/frobt.2021.734548.

Chris Wilson Antuvan, Mark Ison, and Panagiotis Artemiadis. Embedded human control of robots using myoelectric interfaces, 2014. IEEE Transactions on Neural Systems and Rehabilitation Engineering; 10.1109/tnsre.2014.2302212.

Yusuf Aydn, Ozan Tokatl, Volkan Patolu, and aatay Badoan. Stable physical human-robot interaction using fractional order admittance control, 2018. IEEE Transactions on Haptics; 10.1109/toh.2018.2810871.

Kal Backman, Dana Kuli, and Hoam Chung. Reinforcement learning for shared autonomy drone landings, 2023. Autonomous Robots; 10.1007/s10514-023-10143-3.

Ribin Balachandran, Hrishik Mishra, Matteo Cappelli, Bernhard Weber, Cristian Secchi, Christian Ott, and Alin Albu-Schffer. Adaptive authority allocation in shared control of robots using bayesian filters, 2020. openalex; 10.1109/icra40945.2020.9196941.

Roberto Belotti and Karl von Ellenrieder. The effects of switching time on shared human-robot control, 2018. Volume 1: Advances in Control Design Methods; Advances in Nonlinear Control; Advances in Robotics; Assistive and Rehabilitation Robotics; Automotive Dynamics and Emerging Powertrain Technologies; Automotive Systems; Bio Engineering Applications; Bio-Mechatronics and Physical Human Robot Interaction; Biomedical and Neural Systems; Biomedical and Neural Systems Modeling, Diagnostics, and Healthcare; 10.1115/dscc2018-9194.

Federico Benzi, Federica Ferraguti, Giuseppe Riggio, and Cristian Secchi. An energy-based control architecture for shared autonomy, 2022. IEEE Transactions on Robotics; 10.1109/tro.2022.3180885.

Michael Bowman and Xiaoli Zhang. We-filter: Adaptive acceptance criteria for filter-based shared autonomy, 2023. openalex; 10.1109/icra48891.2023.10161228.

Alexander Broad, Ian Abraham, Todd Murphey, and Brenna Argall. Data-driven koopman operators for model-based shared control of human-machine systems, 2020. The International Journal of Robotics Research; 10.1177/0278364920921935.

Samuel Bustamante, Gabriel Quere, Katharina Hagmann, Xuwei Wu, Peter Schmaus, Jrn Vogel, Freek Stulp, and Daniel Leidner. Toward seamless transitions between shared control and supervised autonomy in robotic assistance, 2021. IEEE Robotics and Automation Letters; 10.1109/lra.2021.3064449.

Peng Chang, Rui Luo, Mehrdad Dorostian, and Taskn Padr. A shared control method for collaborative human-robot plug task, 2021. IEEE Robotics and Automation Letters; 10.1109/lra.2021.3098323.

Jieming Chen, Xiang Chen, and Steven Liu. Trajectory planning of autonomous mobile robot using model predictive control in human-robot shared workspace, 2023. 2023 IEEE 3rd International Conference on Electronic Technology, Communication and Information (ICETCI); 10.1109/icetci57876.2023.10176869.

Shaowen Cheng, Yongbin Jin, Yanhong Liang, Lei Jiang, and Hongtao Wang. An efficient grasping shared control architecture for unpredictable and unspecified tasks, 2024. Frontiers in Neurorobotics; 10.3389/fnbot.2024.1429952.

Rahul Chipalkatty, Hannes G. Daepp, Magnus Egerstedt, and Wayne J. Book. Human-in-the-loop: Mpc for shared control of a quadruped rescue robot, 2011. 2011 IEEE/RSJ International Conference on Intelligent Robots and Systems; 10.1109/iros.2011.6094601.

David C. Conner, Stefan Kohlbrecher, Philipp Schillinger, Alberto Romay, Alexander Stumpf, Spyros Maniatopoulos, Hadas Kress-Gazit, and Oskar von Stryk. Collaborative autonomy between high-level behaviors and human operators for control of complex tasks with different humanoid robots, 2018. Springer Tracts in Advanced Robotics The DARPA Robotics Challenge Finals: Humanoid Robots To The Rescue; 10.1007/978-3-319-74666-1_12.

Marco Costanzo, Giuseppe De Maria, and Ciro Natale. Handover control for human-robot and robot-robot collaboration, 2021. Frontiers in Robotics and AI; 10.3389/frobt.2021.672995.

Jacob W. Crandall and Michael A. Goodrich. Characterizing efficiency of human robot interaction: a case study of shared-control teleoperation, 2003. openalex; 10.1109/irds.2002.1043932.

Nicolas Ct, Arnaud Canu, Maroua Bouzid, and Abdel-Iillah Mouaddib. Humans-robots sliding collaboration control in complex environments with adjustable autonomy, 2012. 2012 IEEE/WIC/ACM International Conferences on Web Intelligence and Intelligent Agent Technology; 10.1109/wi-iat.2012.215.

Yuchen Cui, Siddharth Karamcheti, Raj Palleti, Nidhya Shivakumar, Percy Liang, and Dorsa Sadigh. ”no, to the right” – online language corrections for robotic manipulation via shared autonomy, 2023. arXiv; 10.1145/3568162.3578623.

Majid Deldar and Sohel Anwar. Optimal control of a hydrostatic wind turbine drivetrain for efficiency improvements, 2017. Volume 1: Aerospace Applications; Advances in Control Design Methods; Bio Engineering Applications; Advances in Non-Linear Control; Adaptive and Intelligent Systems Control; Advances in Wind Energy Systems; Advances in Robotics; Assistive and Rehabilitation Robotics; Biomedical and Neural Systems Modeling, Diagnostics, and Control; Bio-Mechatronics and Physical Human Robot; Advanced Driver Assistance Systems and Autonomous Vehicles; Automotive Systems; 10.1115/dscc2017-5071.

Xiaoyan Deng, Zhu Liang Yu, Canguang Lin, Zhenghui Gu, and Yuanqing Li. A bayesian shared control approach for wheelchair robot with brain machine interface, 2020. IEEE Transactions on Neural Systems and Rehabilitation Engineering; 10.1109/tnsre.2019.2958076.

Hannah Douglas, Marina Di Vincenzo, Rousslan Fernand Julien Dossa, Luca Nunziante, Shivakanth Sujit, and Kai Arulkumaran. Levels of shared autonomy in brain-robot interfaces: enabling multi-robot multi-human collaboration for activities of daily living, 2026. Frontiers in Human Neuroscience; 10.3389/fnhum.2025.1718713.

Fanny Ficuciello, Guglielmo Tamburrini, Alberto Arezzo, Luigi Villani, and Bruno Siciliano. Autonomy in surgical robots and its meaningful human control, 2018. Paladyn Journal of Behavioral Robotics; 10.1515/pjbr-2019-0002.

Kathleen Fitzsimons, Aleksandra Kalinowska, Julius P Dewald, and Todd D Murphey. Task-based hybrid shared control for training through forceful interaction, 2020. The International Journal of Robotics Research; 10.1177/0278364920933654.

Matthew C. Fontaine and Stefanos Nikolaidis. Evaluating human-robot interaction algorithms in shared autonomy via quality diversity scenario generation, 2022. ACM Transactions on Human-Robot Interaction; 10.1145/3476412.

Paolo Franceschi, Nicola Pedrocchi, and Manuel Beschi. Adaptive impedance controller for human-robot arbitration based on cooperative differential game theory, 2022. 2022 International Conference on Robotics and Automation (ICRA); 10.1109/icra46639.2022.9811853.

Nora Fronemann, Kathrin Pollmann, and Wulf Loh. Should my robot know what’s best for me? human-robot interaction between user experience and ethical design, 2021. AI & Society; 10.1007/s00146-021-01210-3.

Junling Fu, Giorgia Maimone, Elisa Iovene, Jianzhuang Zhao, Alberto Redaelli, Giancarlo Ferrigno, and Elena De Momi. Human-inspired active compliant and passive shared control framework for robotic contact-rich tasks in medical applications, 2025. IEEE Transactions on Robotics; 10.1109/tro.2025.3548493.

Andrea Gaggioli, Alice Chirico, Daniele Di Lernia, Mario A. Maggioni, Clelia Malighetti, Federico Manzi, Antonella Marchetti, Davide Massaro, Francesco Rea, and Domenico Rossignoli. Machines like us and people like you: Toward human-robot shared experience, 2021. Cyberpsychology Behavior and Social Networking; 10.1089/cyber.2021.29216.aga.

Marcus Gardner, C. Sebastian Mancero Castillo, Samuel Wilson, Dario Farina, Etienne Burdet, Boo Cheong Khoo, S. Farokh Atashzar, and Ravi Vaidyanathan. A multimodal intention detection sensor suite for shared autonomy of upper-limb robotic prostheses, 2020. Sensors; 10.3390/s20216097.

S. Gnatzig, F. Schuller, and M. Lienkamp. Human-machine interaction as key technology for driverless driving - a trajectory-based shared autonomy control approach, 2012. 2012 IEEE RO-MAN: The 21st IEEE International Symposium on Robot and Human Interactive Communication; 10.1109/roman.2012.6343867.

Binghan He, Mahsa Ghasemi, Ufuk Topcu, and Luis Sentis. A barrier pair method for safe human-robot shared autonomy, 2021a. arXiv; http://arxiv.org/abs/2112.00279v1.

Binghan He, Mahsa Ghasemi, Ufuk Topcu, and Luis Sentis. A barrier pair method for safe human-robot shared autonomy, 2021b. 2021 60th IEEE Conference on Decision and Control (CDC); 10.1109/cdc45484.2021.9682961.

Daniel Herrera, Flavio Roberti, Ricardo Carelli, Victor Andaluz, Jos Varela, Jessica Ortiz, and Pal Canseco. Modeling and path-following control of a wheelchair in human-shared environments, 2018. International Journal of Humanoid Robotics; 10.1142/s021984361850010x.

Sarah Hopko, Jingkun Wang, and Ranjana Mehta. Human factors considerations and metrics in shared space human-robot collaboration: A systematic review, 2022. Frontiers in Robotics and AI; 10.3389/frobt.2022.799522.

Yoyo Tsung-Yu Hou, Wen-Ying Lee, and Malte Jung. ”should i follow the human, or follow the robot?” - robots in power can have more influence than humans on decision-making, 2023. openalex; 10.1145/3544548.3581066.

Grace Hunt, Sarah Hood, and Tommaso Lenzi. Stand-up, squat, lunge, and walk with a robotic knee and ankle prosthesis under shared neural control, 2021. IEEE Open Journal of Engineering in Medicine and Biology; 10.1109/ojemb.2021.3104261.

Yingxin Huo, Xiang Li, Xuan Zhang, and Dong Sun. Intention-driven variable impedance control for physical human-robot interaction, 2021. openalex; 10.1109/aim46487.2021.9517438.

Ryojun Ikeura and Hikaru Inooka. Variable impedance control of a robot for cooperation with a human, 2002. openalex; 10.1109/robot.1995.525725.

Vahid Izadi and Amir H. Ghasemi. Adaptive impedance control for the haptic shared driving task based on nonlinear mpc, 2020. Volume 1: Adaptive/Intelligent Sys. Control; Driver Assistance/Autonomous Tech.; Control Design Methods; Nonlinear Control; Robotics; Assistive/Rehabilitation Devices; Biomedical/Neural Systems; Building Energy Systems; Connected Vehicle Systems; Control/Estimation of Energy Systems; Control Apps.; Smart Buildings/Microgrids; Education; Human-Robot Systems; Soft Mechatronics/Robotic Components/Systems; Energy/Power Systems; Energy Storage; Estimation/Identification; Vehicle Efficiency/Emissions; 10.1115/dscc2020-3296.

Shail Jadav, Johannes Heidersberger, Christian Ott, and Dongheui Lee. Shared autonomy via variable impedance control and virtual potential fields for encoding human demonstrations *, 2024. 2024 IEEE International Conference on Robotics and Automation (ICRA); 10.1109/icra57147.2024.10610761.

Siddarth Jain and Brenna Argall. Recursive bayesian human intent recognition in shared-control robotics, 2018. openalex; 10.1109/iros.2018.8593766.

Siddarth Jain and Brenna Argall. Probabilistic human intent recognition for shared autonomy in assistive robotics, 2020. ACM Transactions on Human-Robot Interaction; 10.1145/3359614.

Arvid QL Keemink, Herman van der Kooij, and Arno H. A. Stienen. Admittance control for physical human-robot interaction, 2018. The International Journal of Robotics Research; 10.1177/0278364918768950.

Farshad Khadivar, Vincent Mendez, Carolina Correia, Iason Batzianoulis, Aude Billard, and Silvestro Micera. Emg-driven shared human-robot compliant control for in-hand object manipulation in hand prostheses, 2022. Journal of Neural Engineering; 10.1088/1741-2552/aca35f.

Hyomin Kim and Woosung Yang. Variable admittance control based on human-robot collaboration observer using frequency analysis for sensitive and safe interaction, 2021. Sensors; 10.3390/s21051899.

Alistair Knott. Job sharing between human professionals and chatbots: How should 'handovers' happen?, 2023. Journal of AI, Robotics & Workplace Automation; 10.69554/qwwq7223.

Anna Konstant, Nitzan Orr, Michael Hagenow, Isabelle Gundrum, Yu Hen Hu, Bilge Mutlu, Michael Zinn, Michael Gleicher, and Robert G. Radwin. Human-robot collaboration with a corrective shared controlled robot in a sanding task, 2024. Human Factors The Journal of the Human Factors and Ergonomics Society; 10.1177/00187208241272066.

Saroj Kumar and Dayal R. Parhi. Trajectory planning and control of multiple mobile robot using hybrid mkh-fuzzy logic controller, 2022. Robotica; 10.1017/s0263574722000698.

Nicholas Lawrance, Robert DeBortoli, Dylan Jones, Seth McCammon, Lauren Milliken, Austin Nicolai, Thane Somers, and Geoffrey Hollinger. Shared autonomy for low-cost underwater vehicles, 2019. Journal of Field Robotics; 10.1002/rob.21835.

Anni Li, Christos G. Cassandras, and Wei Xiao. Safe optimal interactions between automated and human-driven vehicles in mixed traffic with event-triggered control barrier functions, 2024. 2024 American Control Conference (ACC); 10.23919/acc60939.2024.10644297.

Yanan Li, Keng Peng Tee, Rui Yan, and Shuzhi Sam Ge. Reinforcement learning for human-robot shared control, 2019. Assembly Automation; 10.1108/aa-10-2018-0153.

Yingke Li and Fumin Zhang. Trust-preserved human-robot shared autonomy enabled by bayesian relational event modeling, 2024. IEEE Robotics and Automation Letters; 10.1109/lra.2024.3438040.

Rolif Lima, Somdeb Saha, Vismay Vakharia, Vighnesh Vatsal, and Kaushik Das. Augmenting robot teleoperation with shared autonomy via model predictive control, 2024. 2024 IEEE Conference on Telepresence; 10.1109/telepresence63209.2024.10841639.

Rolif Lima, Somdeb Saha, Nijil George, Vismay Vakharia, Shubham Parab, Sahil Gaonkar, Vighnesh Vatsal, and Kaushik Das. Teleoperated omni-directional dual arm mobile manipulation robotic system with shared control for retail store, 2026. arXiv; 10.1109/smc54092.2024.10831687.

Tsung-Chi Lin, Achyuthan Unni Krishnan, and Zhi Li. The impacts of unreliable autonomy in human-robot collaboration on shared and supervisory control for remote manipulation, 2023. IEEE Robotics and Automation Letters; 10.1109/lra.2023.3287039.

Zuhong Liu, Junhao Ge, Minhao Xiong, Jiahao Gu, Bowei Tang, Wei Jing, and Siheng Chen. It takes two: Learning interactive whole-body control between humanoid robots, 2025. arXiv; http://arxiv.org/abs/2510.10206v1.

Julio S. Lora-Milln, Juan C. Moreno, and Eduardo Rocn. Coordination between partial robotic exoskeletons and human gait: A comprehensive review on control strategies, 2022. Frontiers in Bioengineering and Biotechnology; 10.3389/fbioe.2022.842294.

Dylan P. Losey, Craig G. McDonald, Edoardo Battaglia, and Marcia K. O'Malley. A review of intent detection, arbitration, and communication aspects of shared control for physical human-robot interaction, 2018. Applied Mechanics Reviews; 10.1115/1.4039145.

Zhenyu Lu, Weiyong Si, Ning Wang, and Chenguang Yang. Dynamic movement primitives-based human action prediction and shared control for bilateral robot teleoperation, 2024. IEEE Transactions on Industrial Electronics; 10.1109/tie.2024.3401185.

Shaqi Luo, Min Cheng, Ruqi Ding, Feng Wang, Bing Xu, and Bingkui Chen. Human-robot shared control based on locally weighted intent prediction for a teleoperated hydraulic manipulator system, 2022. IEEE/ASME Transactions on Mechatronics; 10.1109/tmech.2022.3157852.

Jim Mainprice, Rafi Hayne, and Dmitry Berenson. Goal set inverse optimal control and iterative replanning for predicting human reaching motions in shared workspaces, 2016. IEEE Transactions on Robotics; 10.1109/tro.2016.2581216.

Carlo Masone, Antonio Franchi, HH Blthoff, and Paolo Robuffo Giordano. Interactive planning of persistent trajectories for human-assisted navigation of mobile robots, 2012. openalex; 10.1109/iros.2012.6386171.

Kory W. Mathewson and Patrick M. Pilarski. Simultaneous control and human feedback in the training of a robotic agent with actor-critic reinforcement learning, 2016. arXiv; http://arxiv.org/abs/1606.06979v1.

Lin Meng, Catherine Macleod, Bernd Porr, and H. Gollee. Bipedal robotic walking control derived from analysis of human locomotion, 2018. Biological Cybernetics; 10.1007/s00422-018-0750-5.

Kaveh Merat, Jafar Abbaszadeh Chekan, Hassan Salarieh, and Aria Alasty. Online hybrid model predictive controller design for cruise control of automobiles, 2017. Volume 1: Aerospace Applications; Advances in Control Design Methods; Bio Engineering Applications; Advances in Non-Linear Control; Adaptive and Intelligent Systems Control; Advances in Wind Energy Systems; Advances in Robotics; Assistive and Rehabilitation Robotics; Biomedical and Neural Systems Modeling, Diagnostics, and Control; Bio-Mechatronics and Physical Human Robot; Advanced Driver Assistance Systems and Autonomous Vehicles; Automotive Systems; 10.1115/dscc2017-5274.

Jian Mi, Xianbo Zhang, Zhongjie Long, and Jun Wang. A multi-policy rapidly-exploring random tree based mobile robot controller for safe path planning in human-shared environments, 2025. Sensors; 10.3390/s25227008.

Vahidreza Molazadeh, Qiang Zhang, Xuefeng Bao, Brad E. Dicianno, and Nitin Sharma. Shared control of a powered exoskeleton and functional electrical stimulation using iterative learning, 2021. Frontiers in Robotics and AI; 10.3389/frobt.2021.711388.

Dean D. Molinaro, Inseung Kang, and Aaron J. Young. Estimating human joint moments unifies exoskeleton control, reducing user effort, 2024. Science Robotics; 10.1126/scirobotics.adi8852.

Selma Musi and Sandra Hirche. Haptic shared control for human-robot collaboration: A game-theoretical approach, 2020. IFAC-PapersOnLine; 10.1016/j.ifacol.2020.12.2751.

Stefanos Nikolaidis, Yu Xiang Zhu, David Hsu, and Siddhartha Srinivasa. Human-robot mutual adaptation in shared autonomy, 2017. arXiv; 10.1145/2909824.3020252.

Sehoon Oh, Hanseung Woo, and Kyoungchul Kong. Frequency-shaped impedance control for safe human-robot interaction in reference tracking application, 2014. IEEE/ASME Transactions on Mechatronics; 10.1109/tmech.2014.2309118.

Kazuhide Okamoto and Panagiotis Tsiotras. Data-driven human driver lateral control models for developing haptic-shared control advanced driver assist systems, 2019. Robotics and Autonomous Systems; 10.1016/j.robot.2019.01.020.

Shahram Payandeh. Application of shared control strategy in the design of a robotic device, 2001. openalex; 10.1109/acc.2001.945693.

Luka Peternel, Leonel Rozo, Darwin G. Caldwell, and Arash Ajoudani. A method for derivation of robot task-frame control authority from repeated sensory observations, 2017. IEEE Robotics and Automation Letters; 10.1109/lra.2017.2651368.

Jorge Pomares, Ivan Perea, Gabriel J. Garca, Carlos A. Jara, Juan A. Corrales, and Fernando Torres. A multi-sensorial hybrid control for robotic manipulation in human-robot workspaces, 2011. Sensors; 10.3390/s111009839.

Gabriel Quere, Freek Stulp, David Filliat, and Joo Silvrio. A probabilistic approach for learning and adapting shared control skills with the human in the loop, 2024. 2024 IEEE International Conference on Robotics and Automation (ICRA); 10.1109/icra57147.2024.10610956.

Rahaf Rahal, Giulia Matarese, Marco Gabiccini, Alessio Artoni, Domenico Prattichizzo, Paolo Robuffo Giordano, and Claudio Pacchierotti. Caring about the human operator: Haptic shared control for enhanced user comfort in robotic telemanipulation, 2020. IEEE Transactions on Haptics; 10.1109/toh.2020.2969662.

S. M. Mizanoor Rahman. Cyber-physical-social system between a humanoid robot and a virtual human through a shared platform for adaptive agent ecology, 2017. IEEE/CAA Journal of Automatica Sinica; 10.1109/jas.2017.7510760.

S. M. Mizanoor Rahman, Zhanrui Liao, Longsheng Jiang, and Yue Wang. A regret-based autonomy allocation scheme for human-robot shared vision systems in collaborative assembly in manufacturing, 2016. openalex; 10.1109/coase.2016.7743497.

Krishan Rana, Vibhavari Dasagi, Jesse Haviland, Ben Talbot, Michael Milford, and Niko Snderhauf. Bayesian controller fusion: Leveraging control priors in deep reinforcement learning for robotics, 2023. The International Journal of Robotics Research; 10.1177/02783649231167210.

Hao Ren, Li Zhichao, Qingyuan Wu, Xiaodong Ma, and Dan Wu. An adaptive shared control frame and feedback rendering in interactive robot-assisted surgical manipulation, 2025. The International Journal of Medical Robotics and Computer Assisted Surgery; 10.1002/rcs.70069.

Alberto Romay, Stefan Kohlbrecher, Alexander Stumpf, Oskar von Stryk, Spyros Maniatopoulos, Hadas Kress-Gazit, Philipp Schillinger, and David C. Conner. Collaborative autonomy between high-level behaviors and human operators for remote manipulation tasks using different humanoid robots, 2017. Journal of Field Robotics; 10.1002/rob.21671.

Martin J. Rosenstrauch, Tessa J. Pannen, and Jrg Krger. Human robot collaboration - using kinect v2 for iso/ts 15066 speed and separation monitoring, 2018. Procedia CIRP; 10.1016/j.procir.2018.01.026.

Mohammadhadi Sarajchi and Konstantinos Sirlantzis. Evaluating the interaction between human and paediatric robotic lower-limb exoskeleton: a model-based method, 2025. International Journal of Intelligent Robotics and Applications; 10.1007/s41315-025-00421-x.

Daniel Saraphis, Vahid Izadi, and Amirhossein Ghasemi. Towards explainable co-robots: Developing confidence-based shared control paradigms, 2020. Volume 1: Adaptive/Intelligent Sys. Control; Driver Assistance/Autonomous Tech.; Control Design Methods; Nonlinear Control; Robotics; Assistive/Rehabilitation Devices; Biomedical/Neural Systems; Building Energy Systems; Connected Vehicle Systems; Control/Estimation of Energy Systems; Control Apps.; Smart Buildings/Microgrids; Education; Human-Robot Systems; Soft Mechatronics/Robotic Components/Systems; Energy/Power Systems; Energy Storage; Estimation/Identification; Vehicle Efficiency/Emissions; 10.1115/dscc2020-3313.

Shane P. Saunderson and Goldie Nejat. Persuasive robots should avoid authority: The effects of formal and real authority on persuasion in human-robot interaction, 2021. Science Robotics; 10.1126/scirobotics.abd5186.

Flix Schoeller, Mark Miller, Roy Salomon, and Karl Friston. Trust as extended control: Human-machine interactions as active inference, 2021. Frontiers in Systems Neuroscience; 10.3389/fnsys.2021.669810.

Mario Selvaggio, Marco Cognetti, Stefanos Nikolaidis, Serena Ivaldi, and Bruno Siciliano. Autonomy in physical human-robot interaction: A brief survey, 2021. IEEE Robotics and Automation Letters; 10.1109/lra.2021.3100603.

Emmanuel Senft, Sverin Lemaignan, Paul Baxter, Madeleine Bartlett, and Tony Belpaeme. Teaching robots social autonomy from in situ human guidance, 2019. Science Robotics; 10.1126/scirobotics.aat1186.

Noel Sharkey. Towards a principle for the human supervisory control of robot weapons, 2014. Politica & Societ; 10.4476/77105.

Reid Simmons, Sanjiv Singh, Frederik W. Heger, Laura M. Hiatt, Seth Koterba, Nik A. Melchior, and Brennan Sellner. Human-robot teams for large-scale assembly, 2018. Figshare; 10.1184/r1/6555074.v1.

Justin Storms and Dawn M. Tilbury. Blending of human and obstacle avoidance control for a high speed mobile robot, 2014. openalex; 10.1109/acc.2014.6859352.

Fabio Stroppa, Mario Selvaggio, Nathaniel Agharese, MingLuo, Laura H. Blumenschein, Elliot W. Hawkes, and Allison M. Okamura. Shared-control teleoperation paradigms on a soft growing robot manipulator, 2021. arXiv; 10.1007/s10846-023-01919-x.

Hang Su, Wen Qi, Yunus Schmirander, Salih Ertug Ovur, Shuting Cai, and Xiaoming Xiong. A human activity-aware shared control solution for medical human-robot interaction, 2022. Assembly Automation; 10.1108/aa-12-2021-0174.

Yuzhu Sun, Mien Van, Stephen McIlvanna, Nhat Nguyen Minh, Sen McLoone, and Dariusz Ceglarek. Adaptive admittance control for safety-critical physical human robot collaboration, 2023. IFAC-PapersOnLine; 10.1016/j.ifacol.2023.10.1772.

Gokul Swamy, Siddharth Reddy, Sergey Levine, and Anca D. Dragan. Scaled autonomy: Enabling human operators to control robot fleets, 2019. arXiv; http://arxiv.org/abs/1910.02910v2.

Nazish Tahir and Ramviyas Parasuraman. Analog twin framework for human and ai supervisory control and teleoperation of robots, 2022. IEEE Transactions on Systems Man and Cybernetics Systems; 10.1109/tsmc.2022.3216206.

C. Ton, Zhen Kan, and S. S. Mehta. Obstacle avoidance control of a human-in-the-loop mobile robot system using harmonic potential fields, 2017. Robotica; 10.1017/s0263574717000510.

Cristina Urdiales, Alberto Poncela, Isabel Snchez-Tato, Francesco Galluppi, M. Olivetti, and F. Sandoval. Efficiency based reactive shared control for collaborative human/robot navigation, 2007. openalex; 10.1109/iros.2007.4399288.

Wietse van Dijk, Saskia J. Baltrusch, Ezra Dessers, and Michiel P. de Looze. The effect of human autonomy and robot work pace on perceived workload in human-robot collaborative assembly work, 2023. Frontiers in Robotics and AI; 10.3389/frobt.2023.1244656.

Frdric Vanderhaegen. Heuristic-based method for conflict discovery of shared control between humans and autonomous systems - a driving automation case study, 2021. Robotics and Autonomous Systems; 10.1016/j.robot.2021.103867.

Cristian Alejandro Vergara, Gianni Borghesan, Erwin Aertbelin, and Joris De Schutter. Incorporating artificial skin signals in the constraint-based reactive control of human-robot collaborative manipulation tasks, 2019. Industrial Robot: the international journal of robotics research and application; 10.1108/ir-08-2018-0172.

Ruben Verhagen, Mark A. Neerincx, and Myrthe L. Tielman. Meaningful human control and variable autonomy in human-robot teams for firefighting, 2024. Frontiers in Robotics and AI; 10.3389/frobt.2024.1323980.

Xiaoyu Wang, Alireza Haji Fathaliyan, and Veronica J. Santos. Toward shared autonomy control schemes for human-robot systems: Action primitive recognition using eye gaze features, 2020. Frontiers in Neurorobotics; 10.3389/fnbot.2020.567571.

Yan Wei, Mingshuang Hao, Kehong Jin, Xinyi Yu, and Linlin Ou. Output-constrained adaptive optimal control for physical human-robot interaction using electromyography signals, 2025. Asian Journal of Control; 10.1002/asjc.70001.

Shiyu Xu, Scott D. Moore, and Akansel Cosgun. Shared-control robotic manipulation in virtual reality, 2022. 2022 International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA); 10.1109/hora55278.2022.9800046.

Haotian Xue, Youssef Michel, and Dongheui Lee. A shared control approach based on first-order dynamical systems and closed-loop variable stiffness control, 2023. Journal of Intelligent & Robotic Systems; 10.1007/s10846-023-02023-w.

Xinbo Yu, Bin Li, Wei He, Yanghe Feng, Long Cheng, and Carlos Silvestre. Adaptive-constrained impedance control for human-robot co-transportation, 2021. IEEE Transactions on Cybernetics; 10.1109/tcyb.2021.3107357.

Dandan Zhang, Zicong Wu, Junhong Chen, Ruiqi Zhu, Adnan Munawar, Bo Xiao, Yuan Guan, Hang Su, Wuzhou Hong, and Yao Guo. Human-robot shared control for surgical robot based on context-aware sim-to-real adaptation, 2022a. 2022 International Conference on Robotics and Automation (ICRA); 10.1109/icra46639.2022.9812379.

Dianhao Zhang, Mien Van, Stephen Mcllvanna, Yuzhu Sun, and Sen McLoone. Adaptive safety-critical control with uncertainty estimation for human-robot collaboration, 2023. IEEE Transactions on Automation Science and Engineering; 10.1109/tase.2023.3320873.

Jun Zhang, Xuanchen Zhang, Yong Cheng, Yang Cheng, Qiong Zhang, and Kun Lu. Nonlinear model predictive control (nmpc) based shared autonomy for bilateral teleoperation in cfetr remote handling, 2024. Nuclear Engineering and Technology; 10.1016/j.net.2024.06.005.

Pengfei Zhang, Xueshan Gao, Mingda Miao, and Peng Zhao. Design and control of a lower limb rehabilitation robot based on human motion intention recognition with multi-source sensor information, 2022b. Machines; 10.3390/machines10121125.

Hao Zhou, Xin Zhang, and Jinguo Liu. A corrective shared control architecture for human-robot collaborative polishing tasks, 2024. Robotics and Computer-Integrated Manufacturing; 10.1016/j.rcim.2024.102876.

Yuanchao Zhu, Canjun Yang, Qianxiao Wei, Xin Wu, and Wei Yang. Human-robot shared control for humanoid manipulator trajectory planning, 2020. Industrial Robot: the international journal of robotics research and application; 10.1108/ir-10-2019-0217.

Katie Z. Zhuang, Nicolas Sommer, Vincent Mendez, Saurav Aryan, Emanuele Formento, Edoardo D'Anna, Fiorenzo Artoni, Francesco Petrini, Giuseppe Granata, and Giovanni Cannaviello. Shared human-robot proportional control of a dexterous myoelectric prosthesis, 2019. Nature Machine Intelligence; 10.1038/s42256-019-0093-5.

Mark Zolotas, Murphy Wonsick, Philip Long, and Takn Padr. Motion polytopes in virtual reality for shared control in remote manipulation applications, 2021. Frontiers in Robotics and AI; 10.3389/frobt.2021.730433.